

(12) **United States Patent**
Chen et al.

(10) **Patent No.:** **US 12,406,876 B2**
(45) **Date of Patent:** **Sep. 2, 2025**

(54) **SEMICONDUCTOR STRUCTURE INCLUDING ISOLATION STRUCTURE WITH DIELECTRIC STACK AND METHOD OF FORMING THE SAME**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 728 days.

(21) Appl. No.: **17/587,299**

(22) Filed: **Jan. 28, 2022**

(65) **Prior Publication Data**
US 2023/0245921 A1 Aug. 3, 2023

(51) **Int. Cl.**
H01L 21/762 (2006.01)
H10D 84/01 (2025.01)
(Continued)

(52) **U.S. Cl.**
CPC ... **H01L 21/76224** (2013.01); **H10D 84/0151** (2025.01); **H10D 84/038** (2025.01); **H10D 84/834** (2025.01); **H10D 84/0149** (2025.01)

(58) **Field of Classification Search**
CPC .. H10D 30/024; H10D 30/0243; H10D 30/62; H10D 30/6211; H10D 30/6212; H10D 30/6213; H10D 30/6215; H10D 30/6217; H10D 30/6218; H10D 30/6219; H10D

62/115; H10D 62/116; H10D 64/017; H10D 84/0151; H10D 84/0153; H10D 84/0158; H10D 84/834; H10D 84/853; H10D 84/0193; H10D 84/0188; H01L 21/76213; H01L 21/76216; H01L 21/76218; H01L 21/76221; H01L 21/76224; H01L 21/76227; H01L 21/76229; H01L 21/76232; H01L 21/76235; H01L 21/76237; H01L 21/7624; H01L 21/76234; H01L 21/76245; H01L 21/76251; H01L 21/76248; H01L 21/76254; H01L 21/76256; H01L 21/76259; H01L 21/76262;

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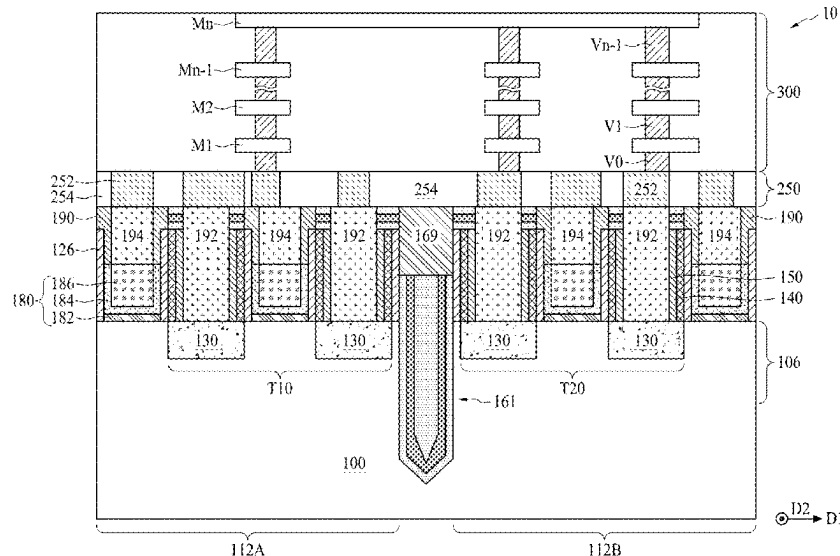
Primary Examiner — Eric A. Ward

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(57) **ABSTRACT**

The present disclosure provides a method of manufacturing a semiconductor structure. The method includes providing a substrate; forming a fin structure on the substrate; forming a first dummy gate on the fin structure; forming a trench to penetrate the first dummy gate and the fin structure; forming a dielectric stack in the trench; removing a top portion of the dielectric stack in the trench to leave a lower portion of the dielectric stack in the trench; and forming a protective layer in the trench and directly on the lower portion of the dielectric stack.

20 Claims, 40 Drawing Sheets



(51) **Int. Cl.**

H10D 84/03 (2025.01)

H10D 84/83 (2025.01)

(58) **Field of Classification Search**

CPC H01L 21/76264; H01L 21/7627; H01L
21/76291; H01L 21/76289; H01L
21/76286; H01L 21/76283; H01L
21/76281; H01L 21/76278; H01L
21/76275; H01L 21/76272; H01L
21/76297; H01L 21/76294; H01L 21/765;
H01L 21/764; H01L 21/763; H01L
21/762; H01L 21/76202; H01L 21/76205;
H01L 21/76208; H01L 21/7621

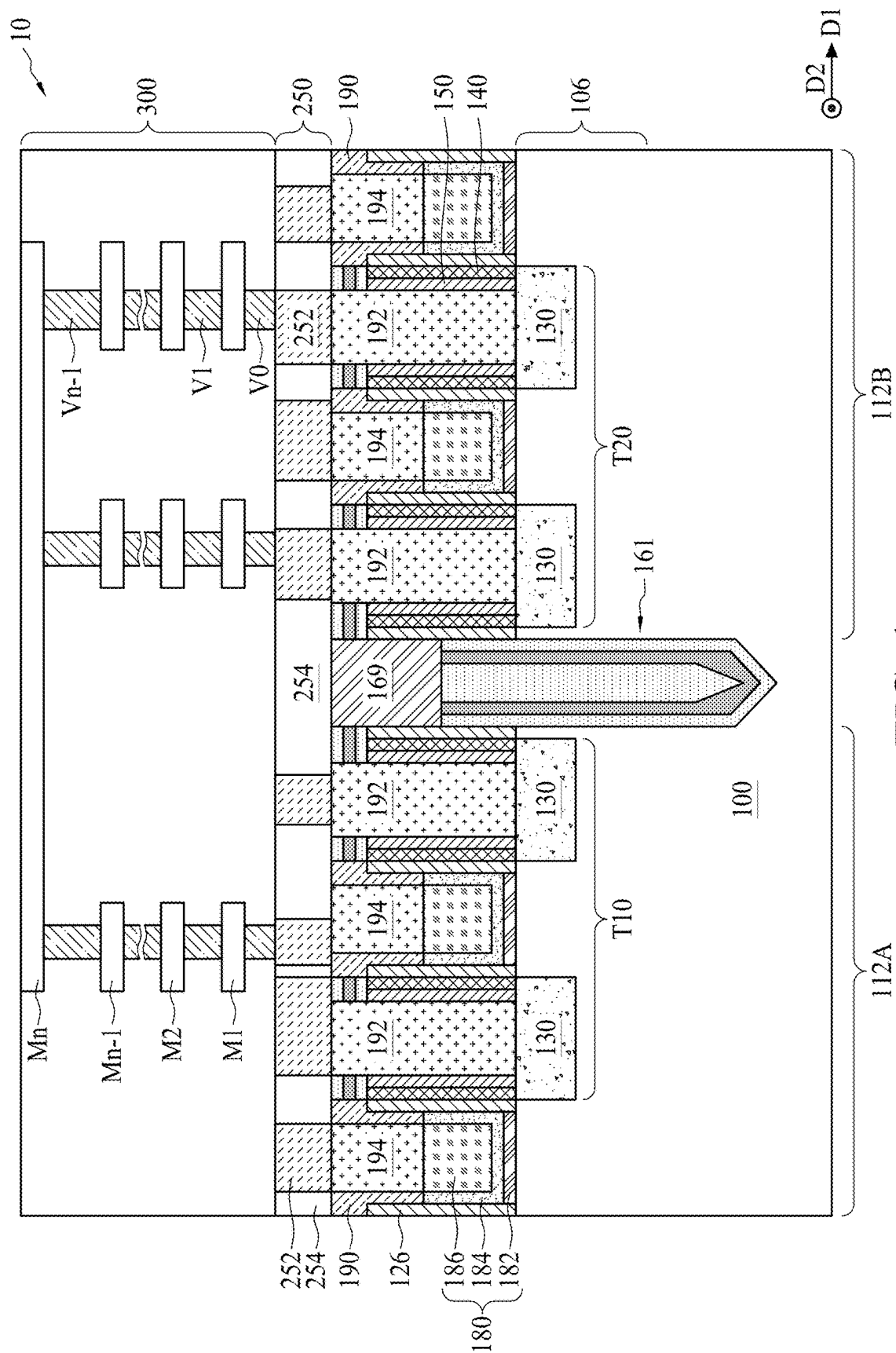
See application file for complete search history.

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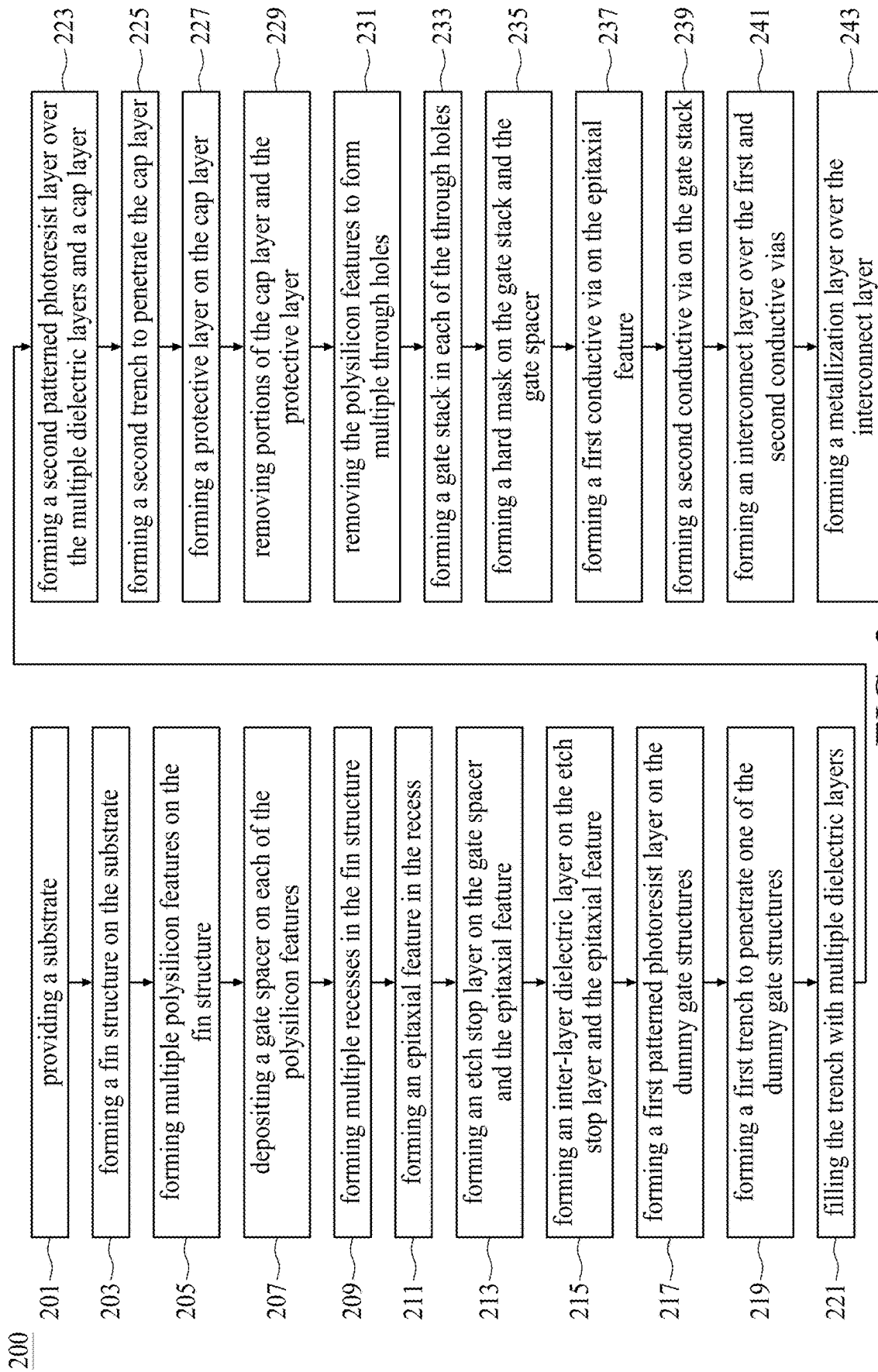


FIG. 2

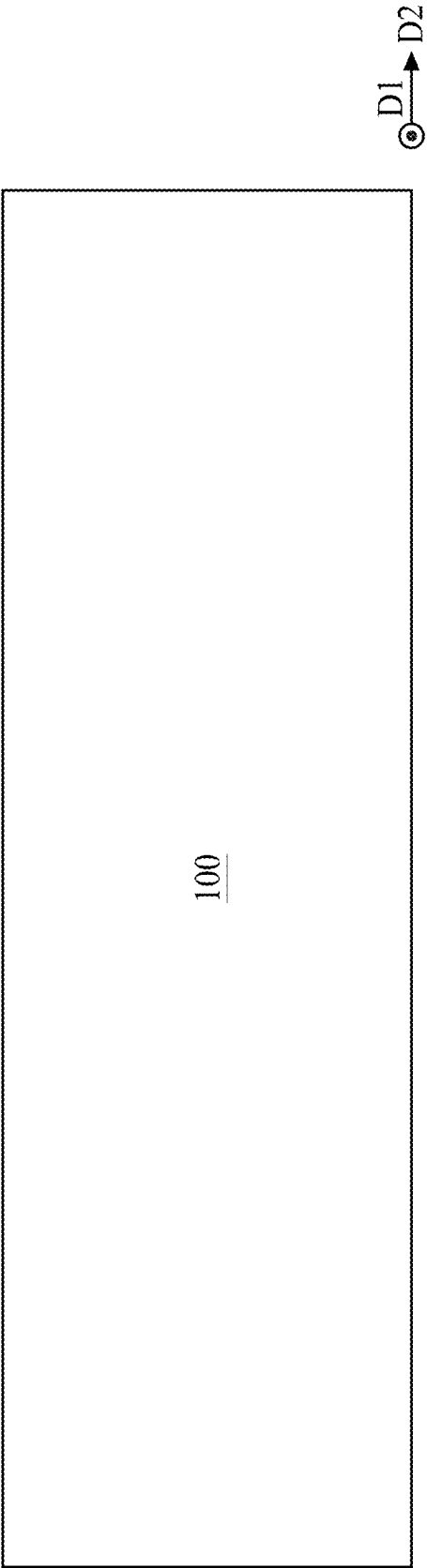


FIG. 3

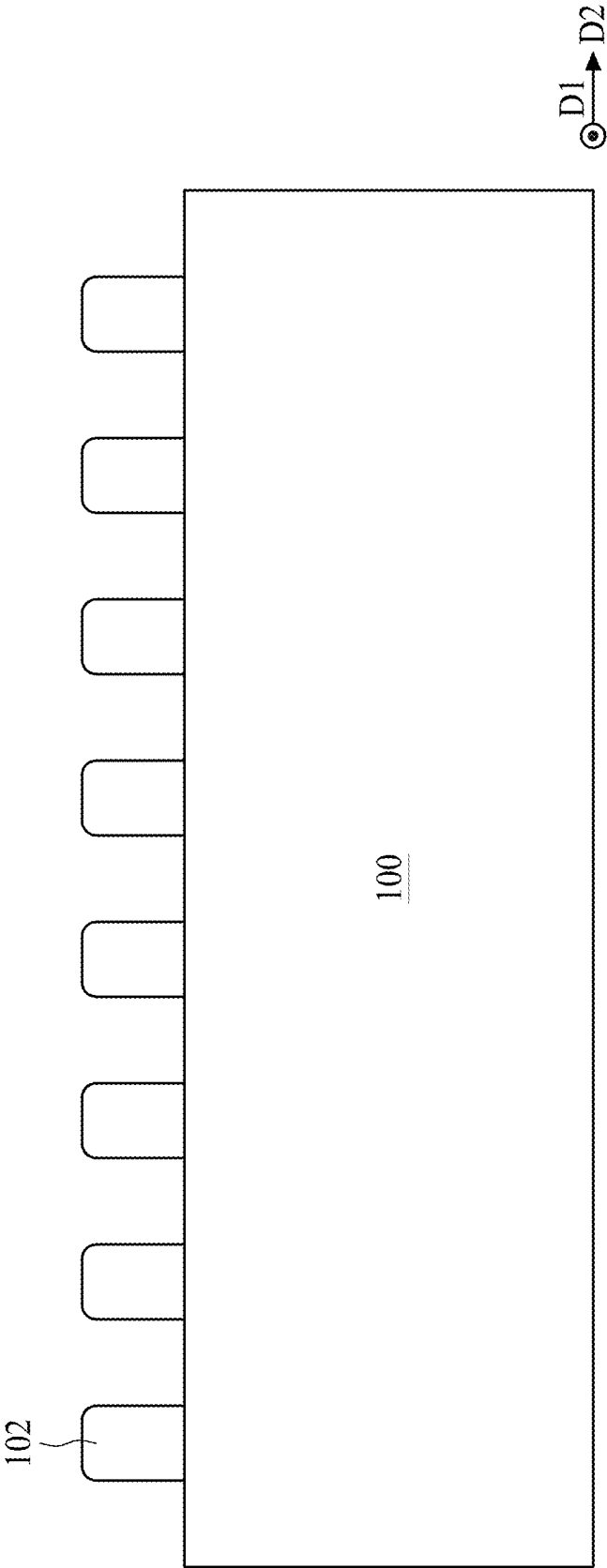


FIG. 4A

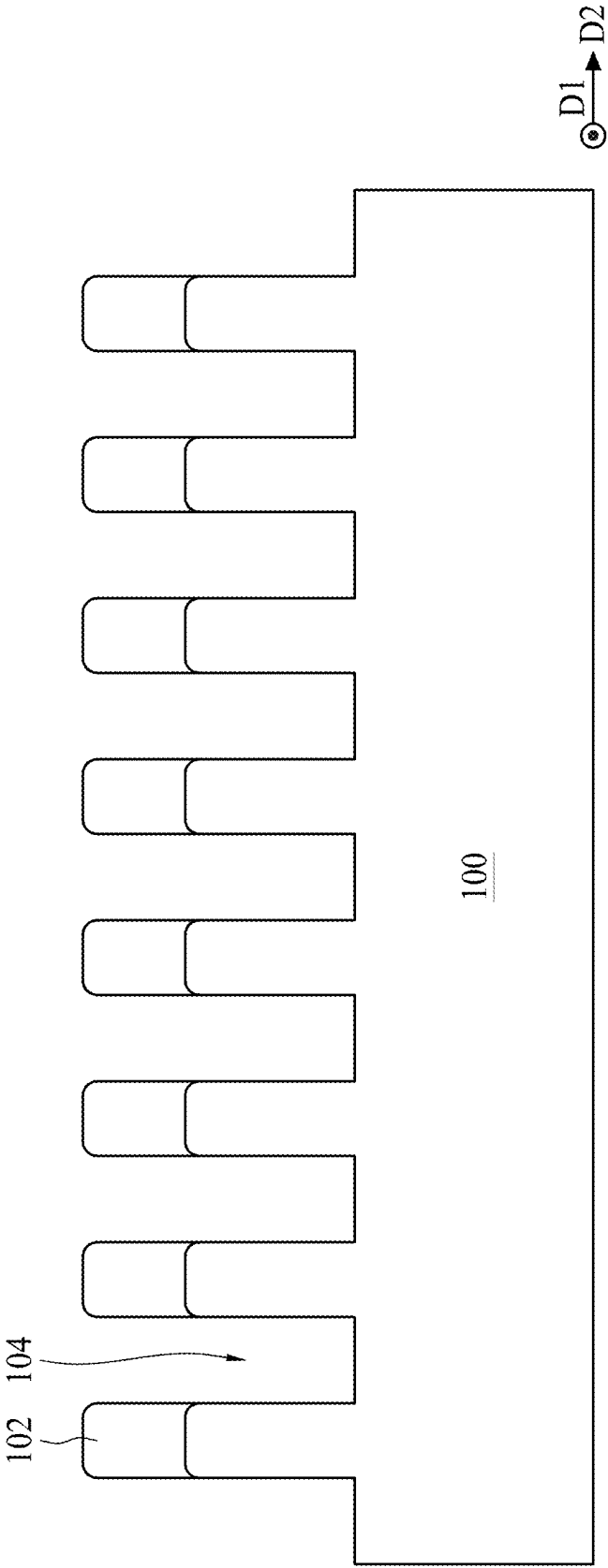


FIG. 4B

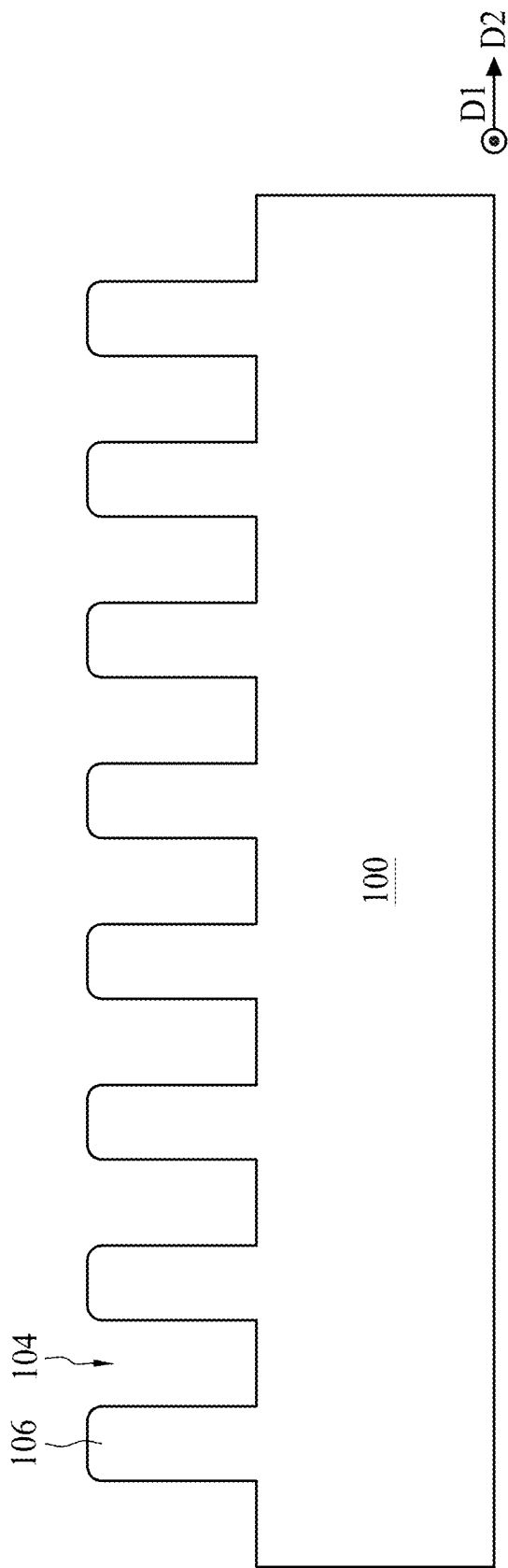


FIG. 4C

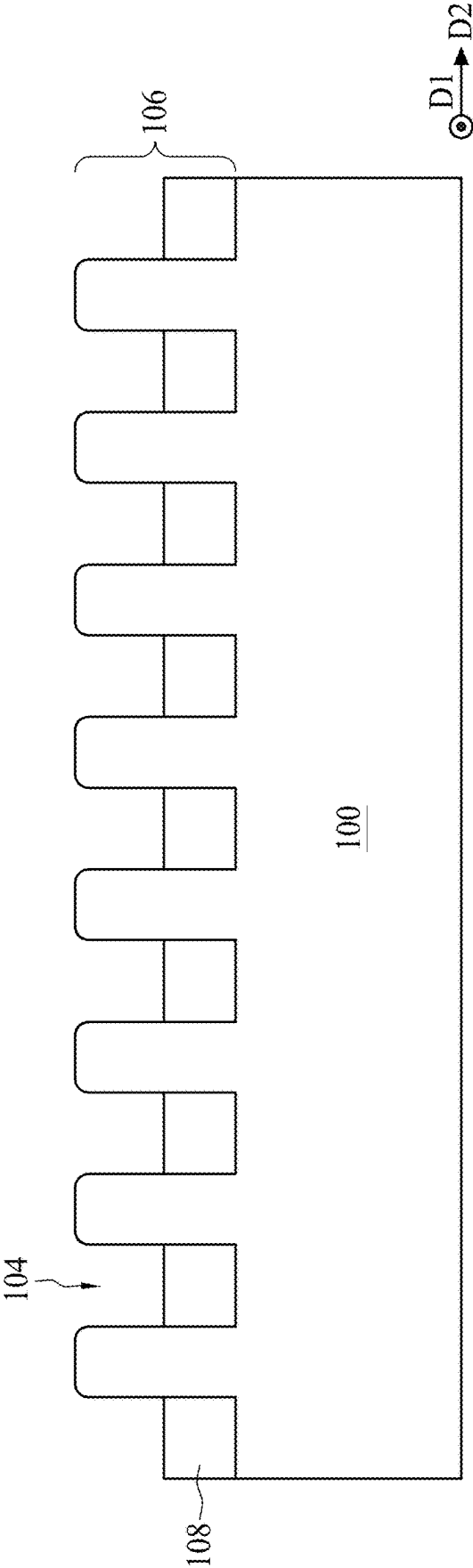


FIG. 4D

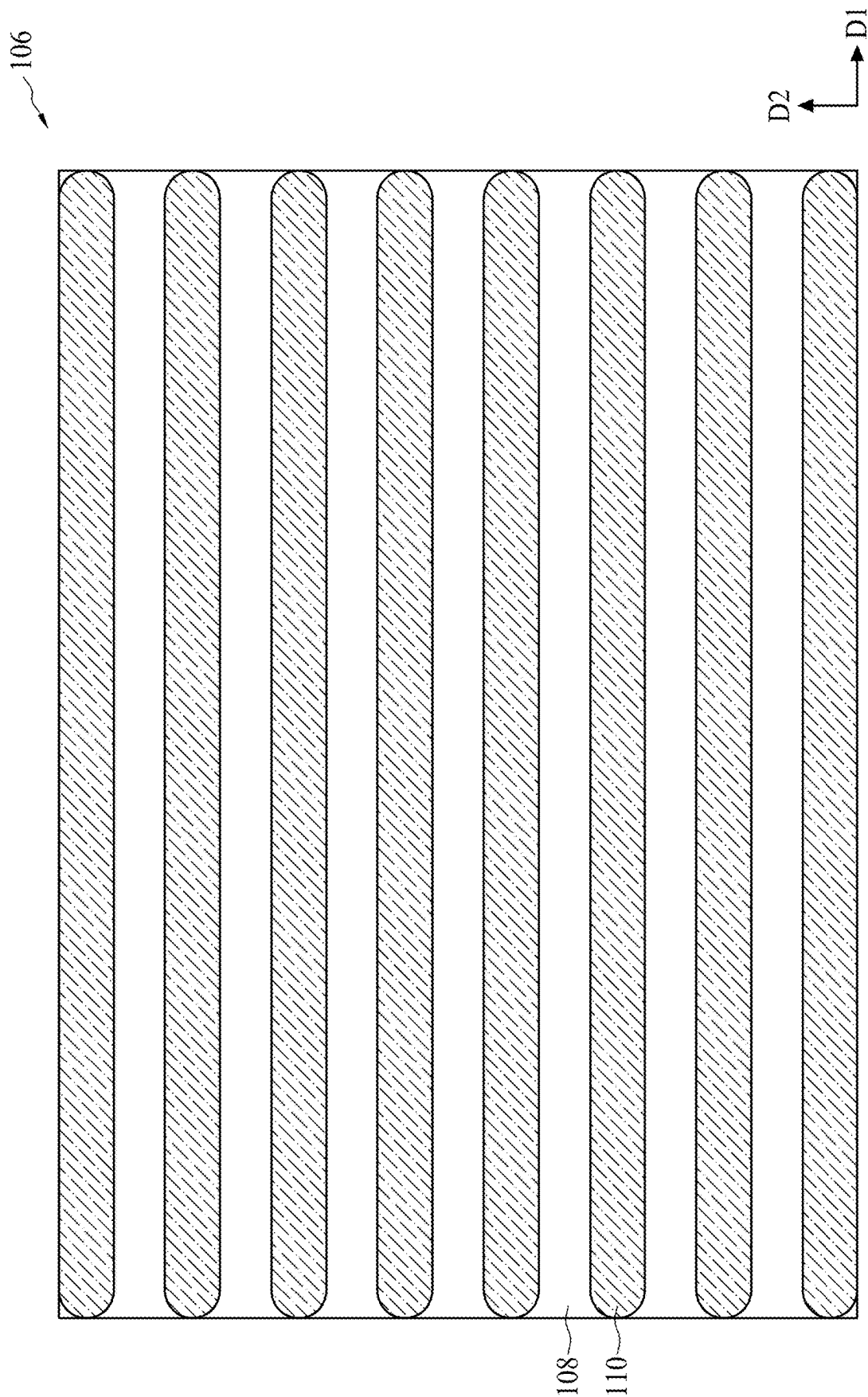


FIG. 4E

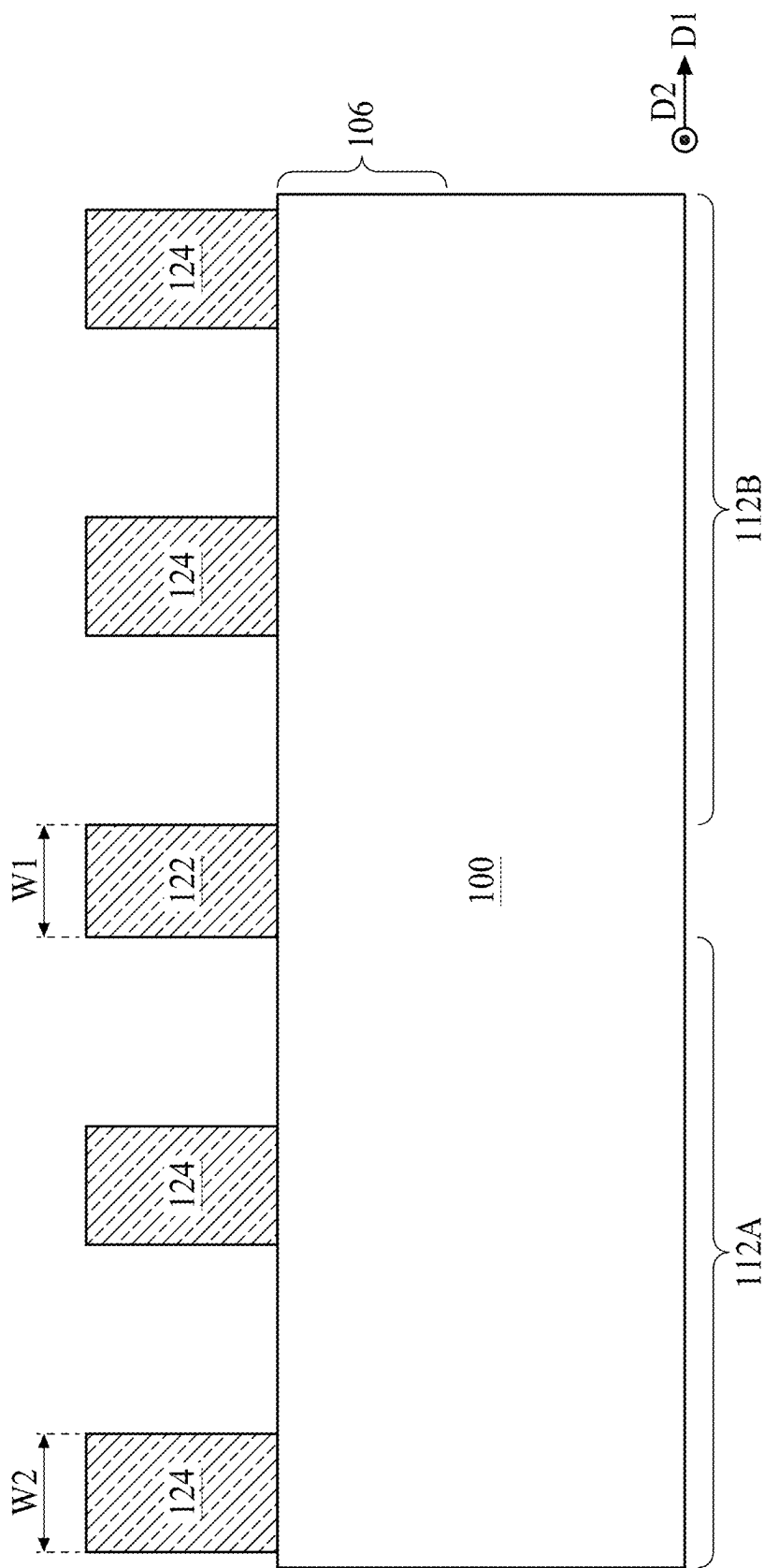


FIG. 5A

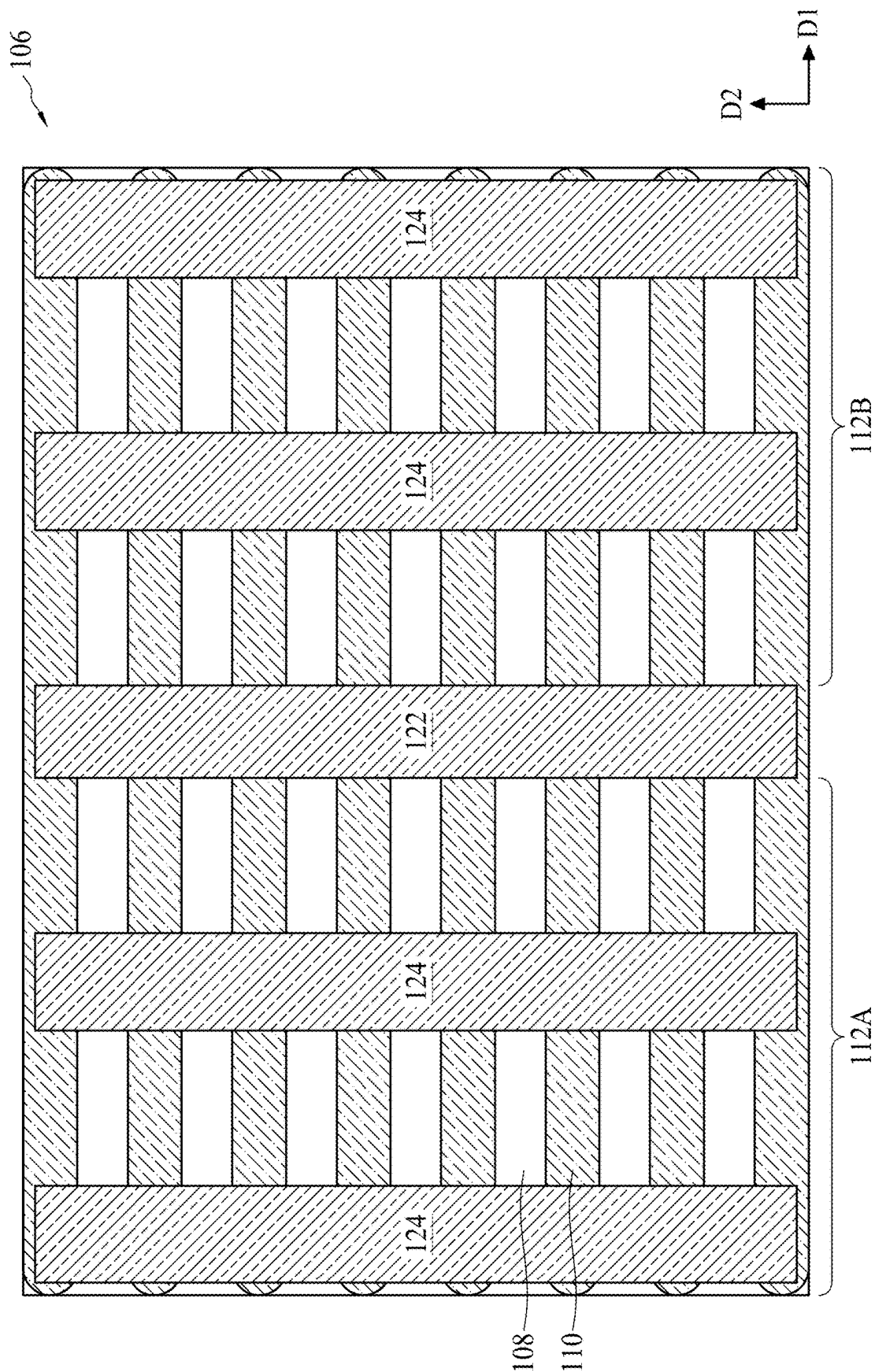
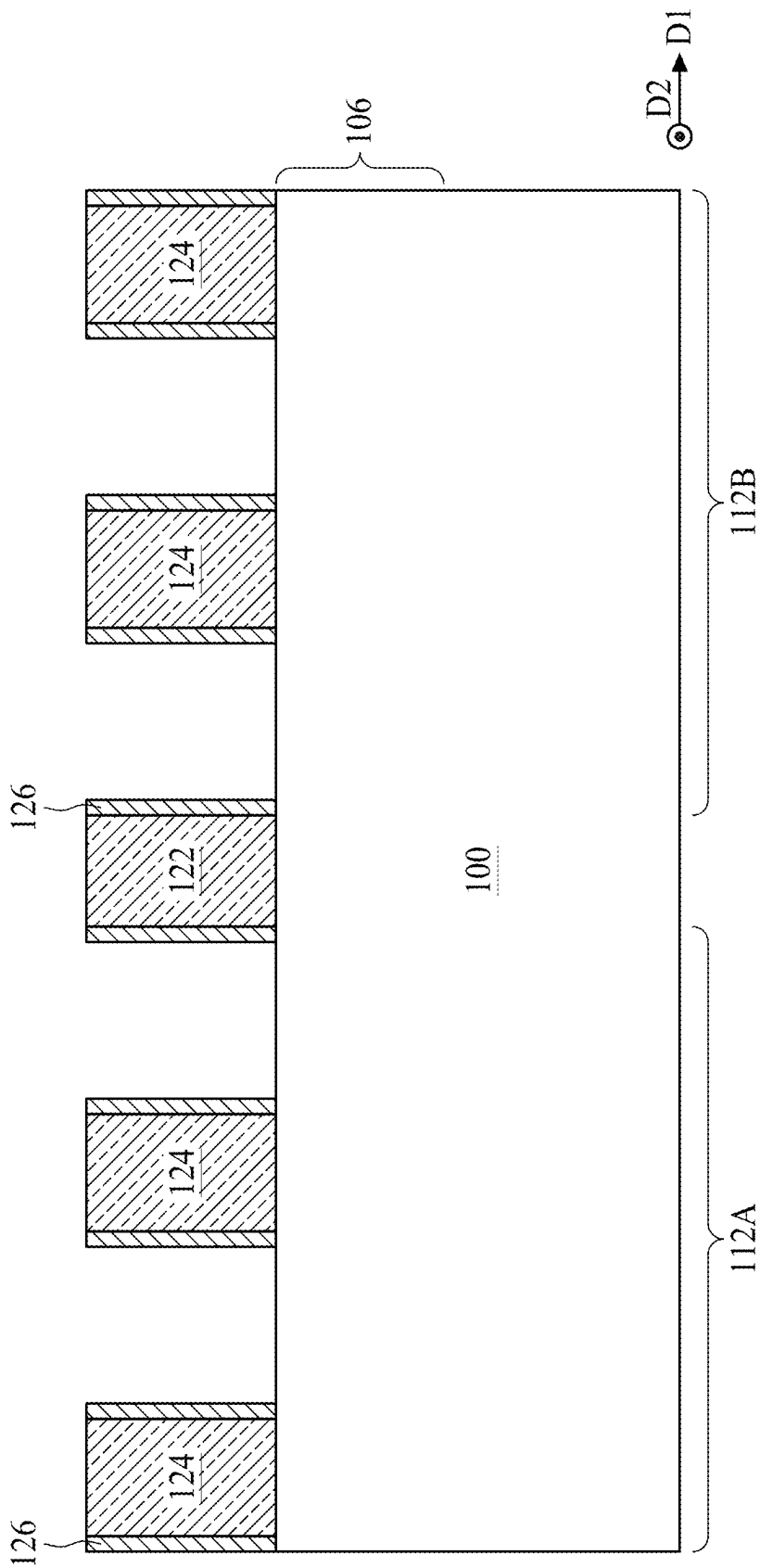


FIG. 5B



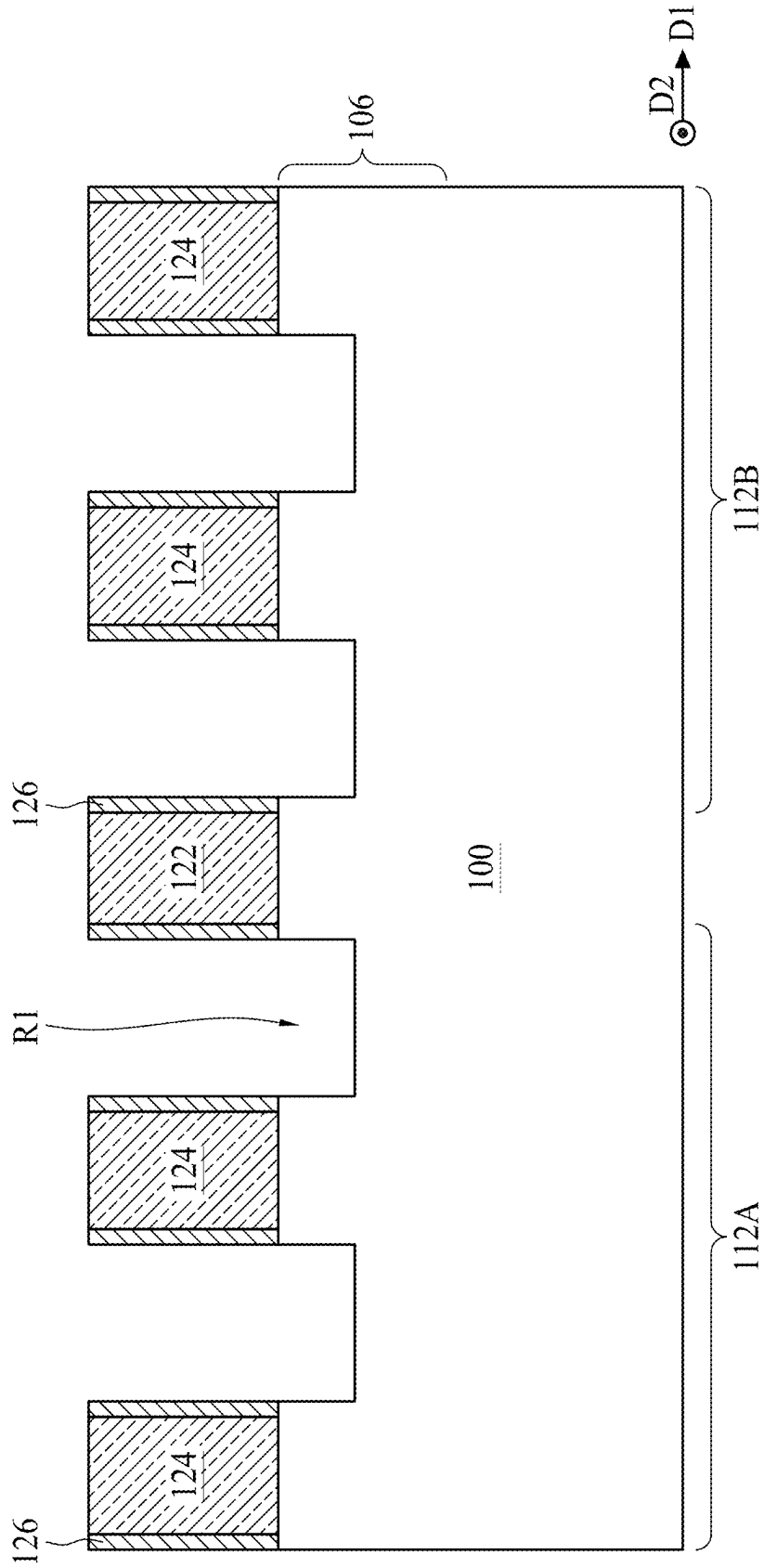
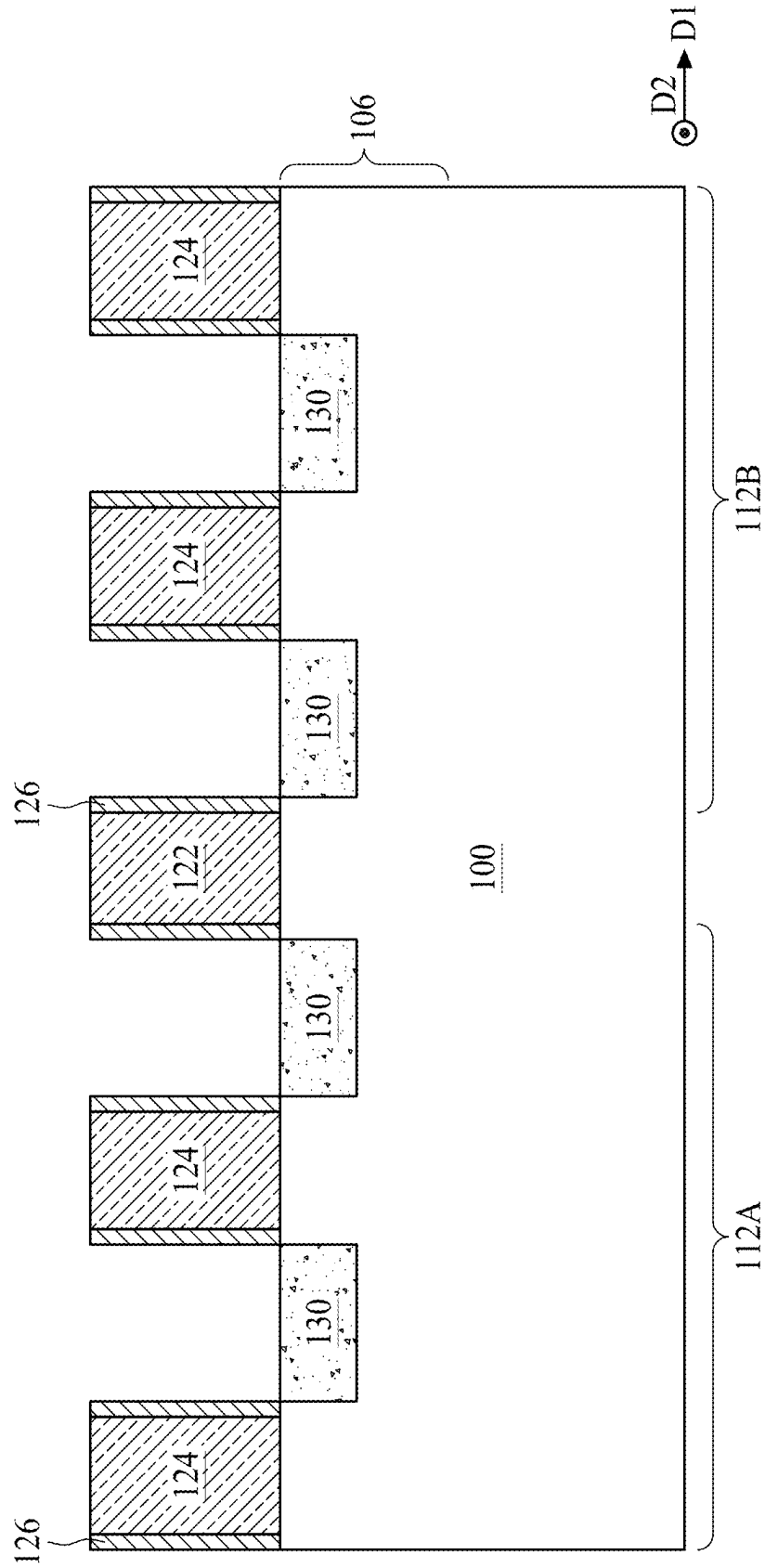
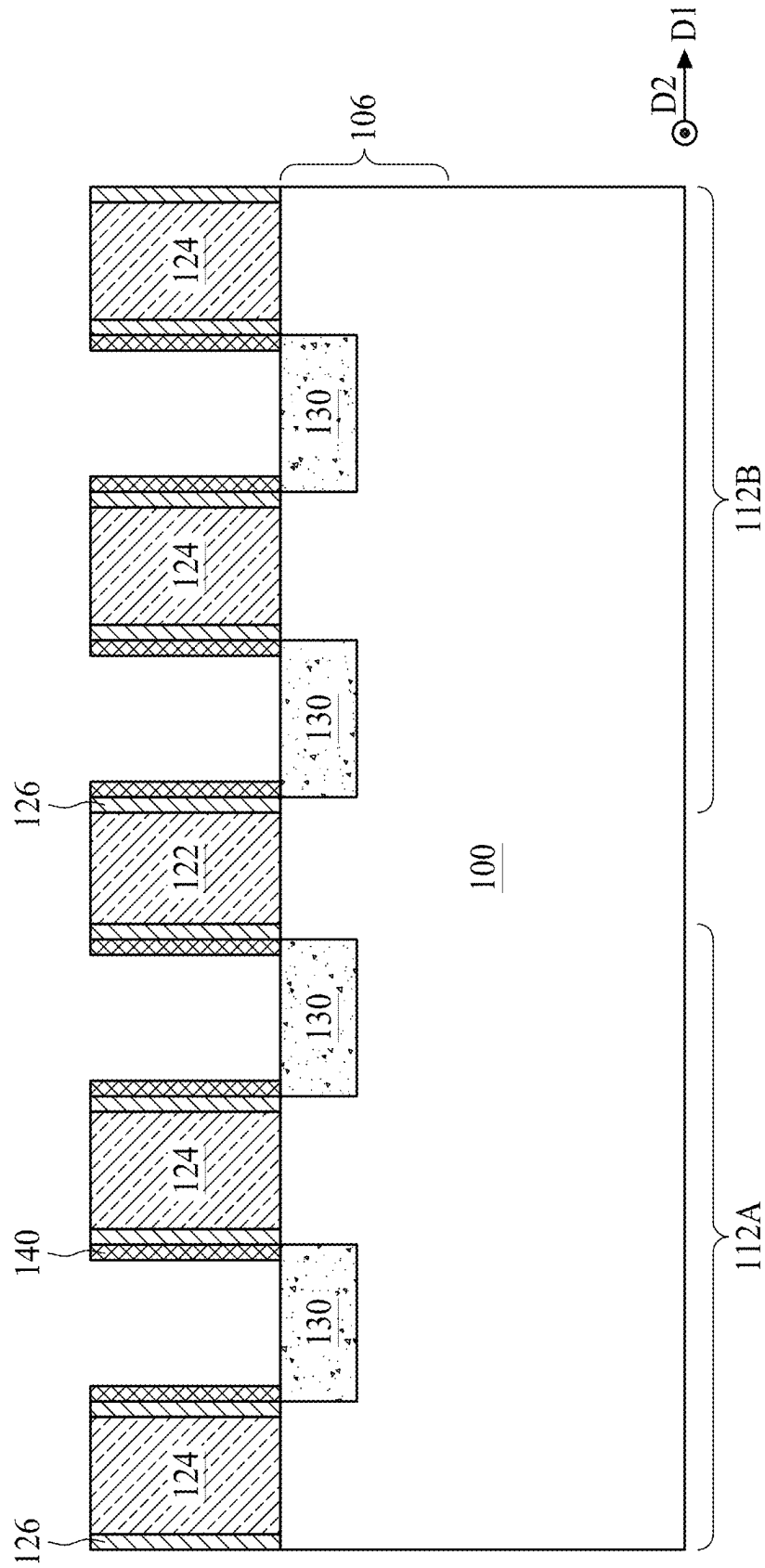


FIG. 7





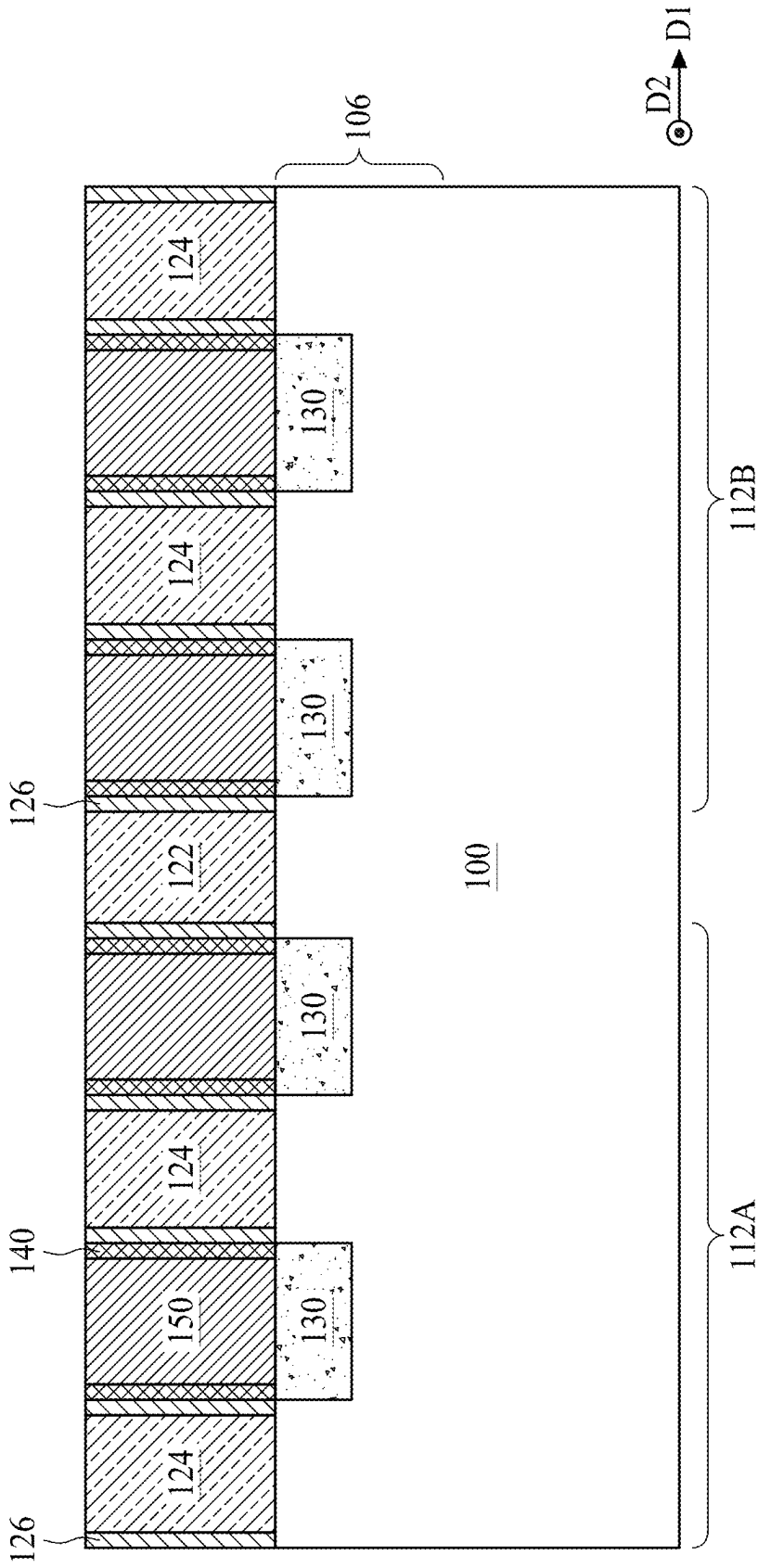
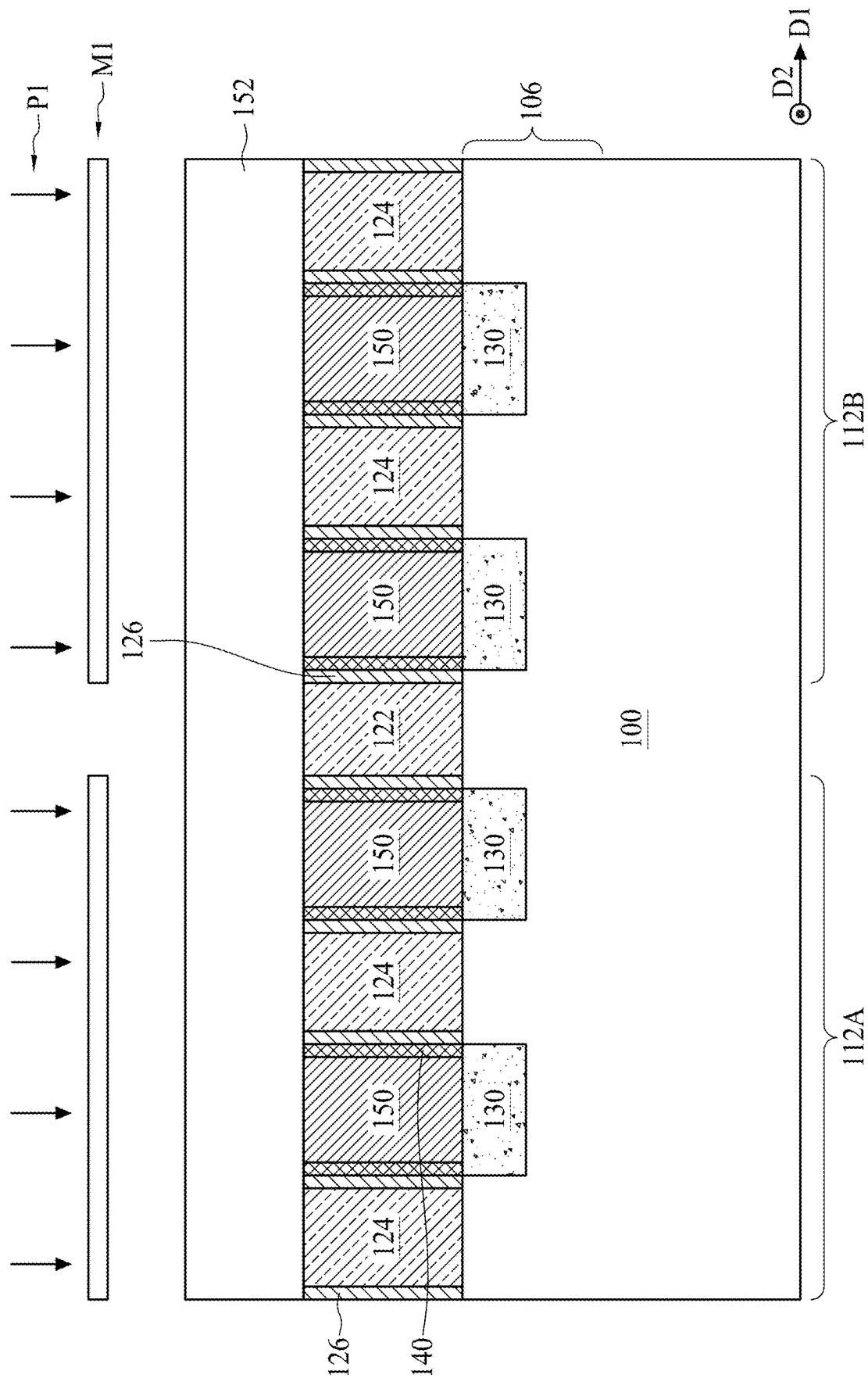


FIG. 10



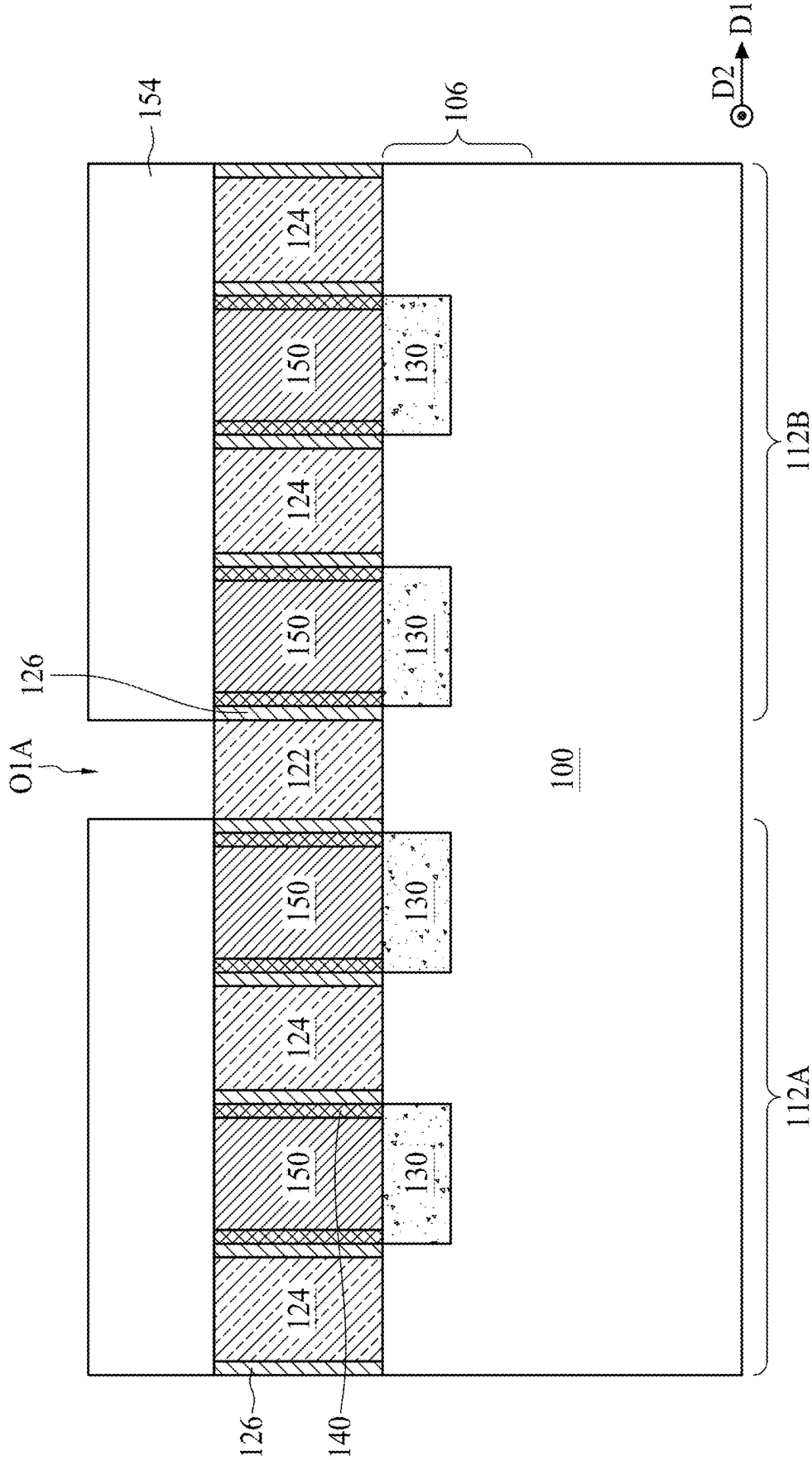


FIG. 11B

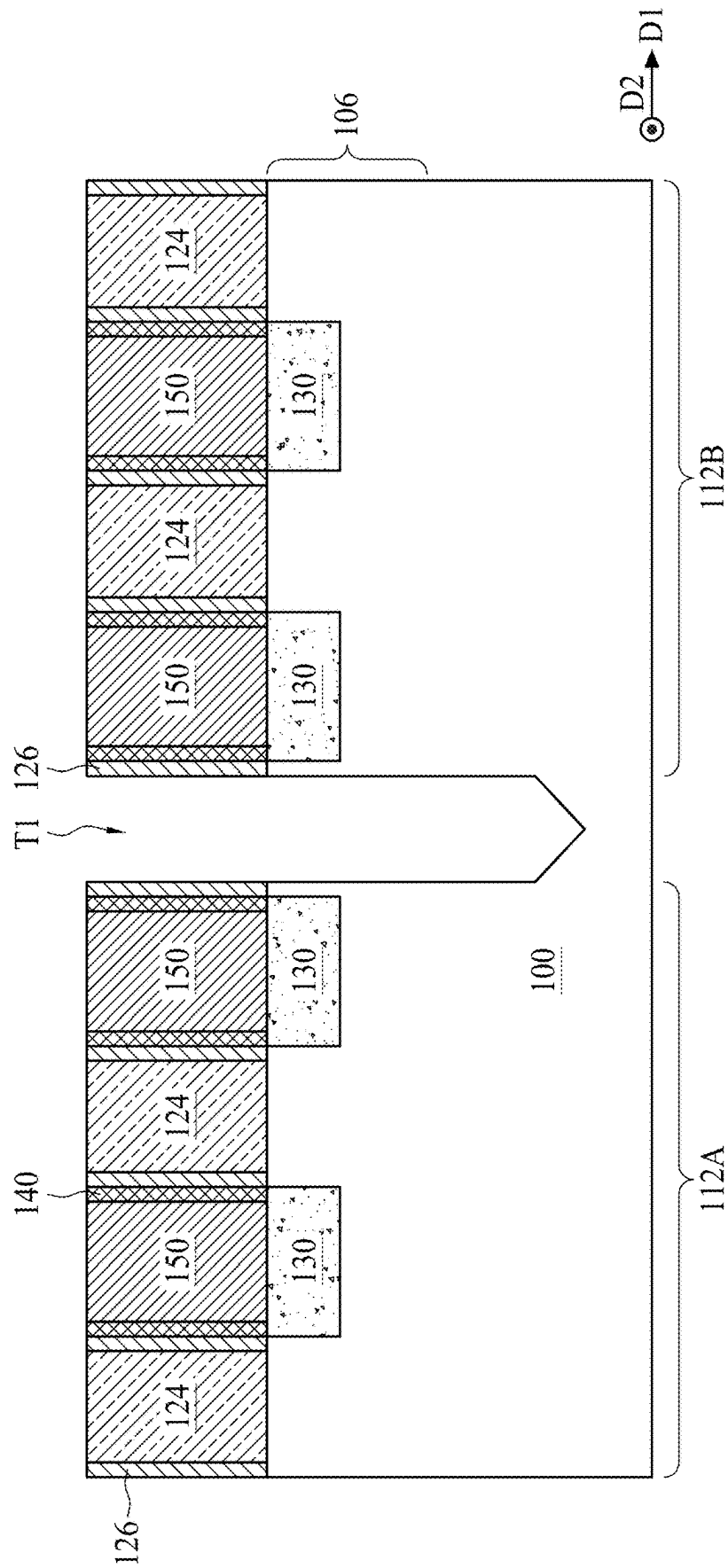


FIG. 12A

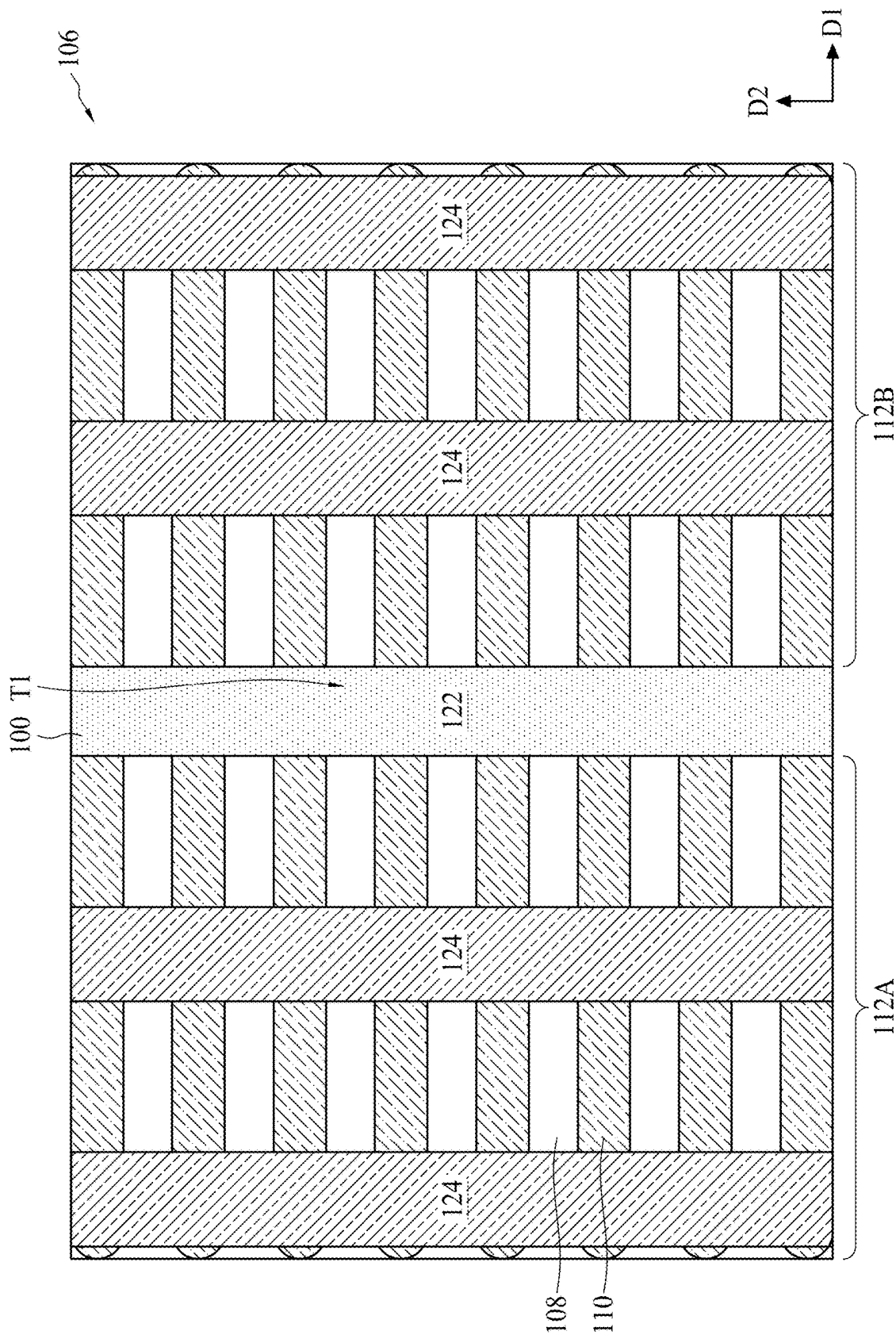


FIG. 12B

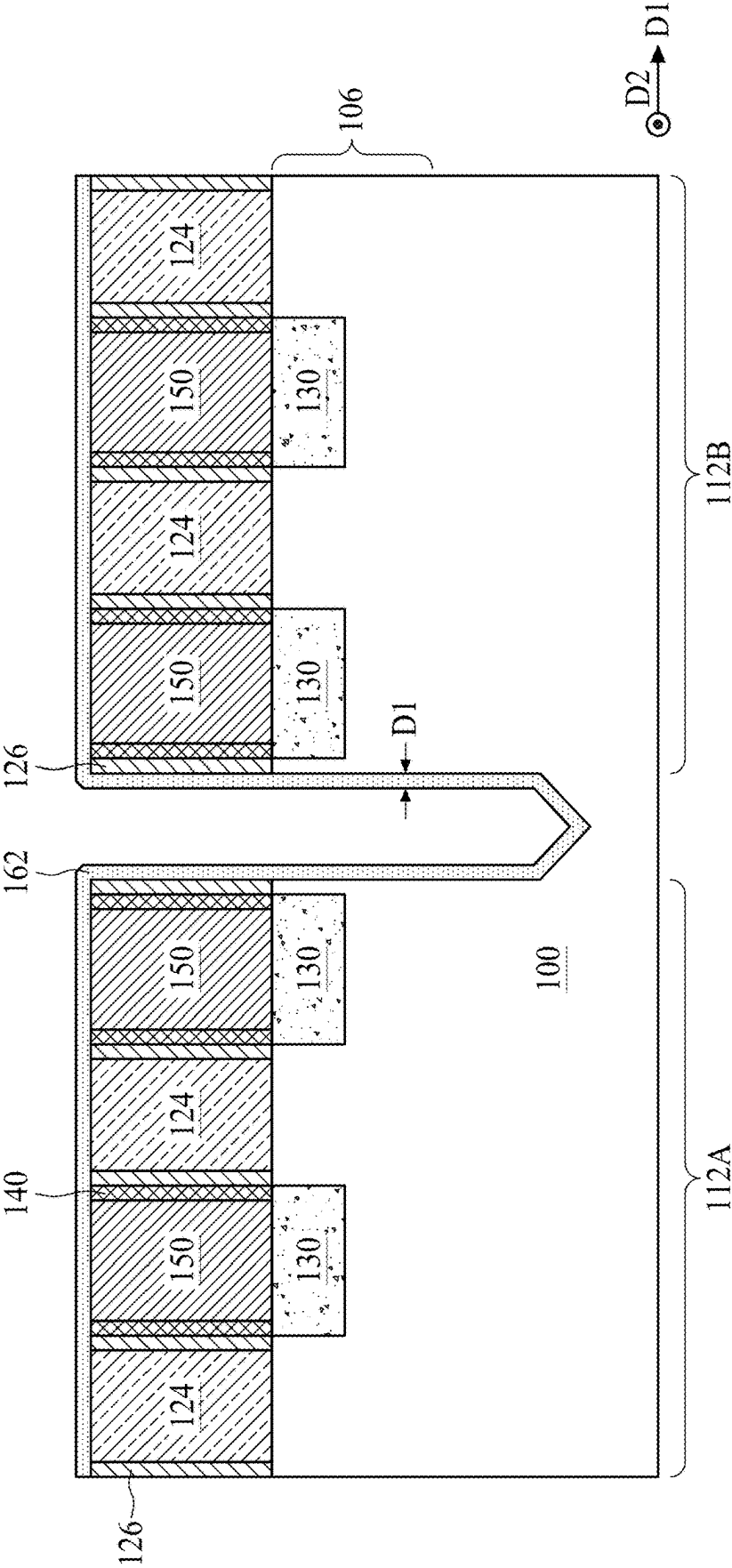


FIG. 13A

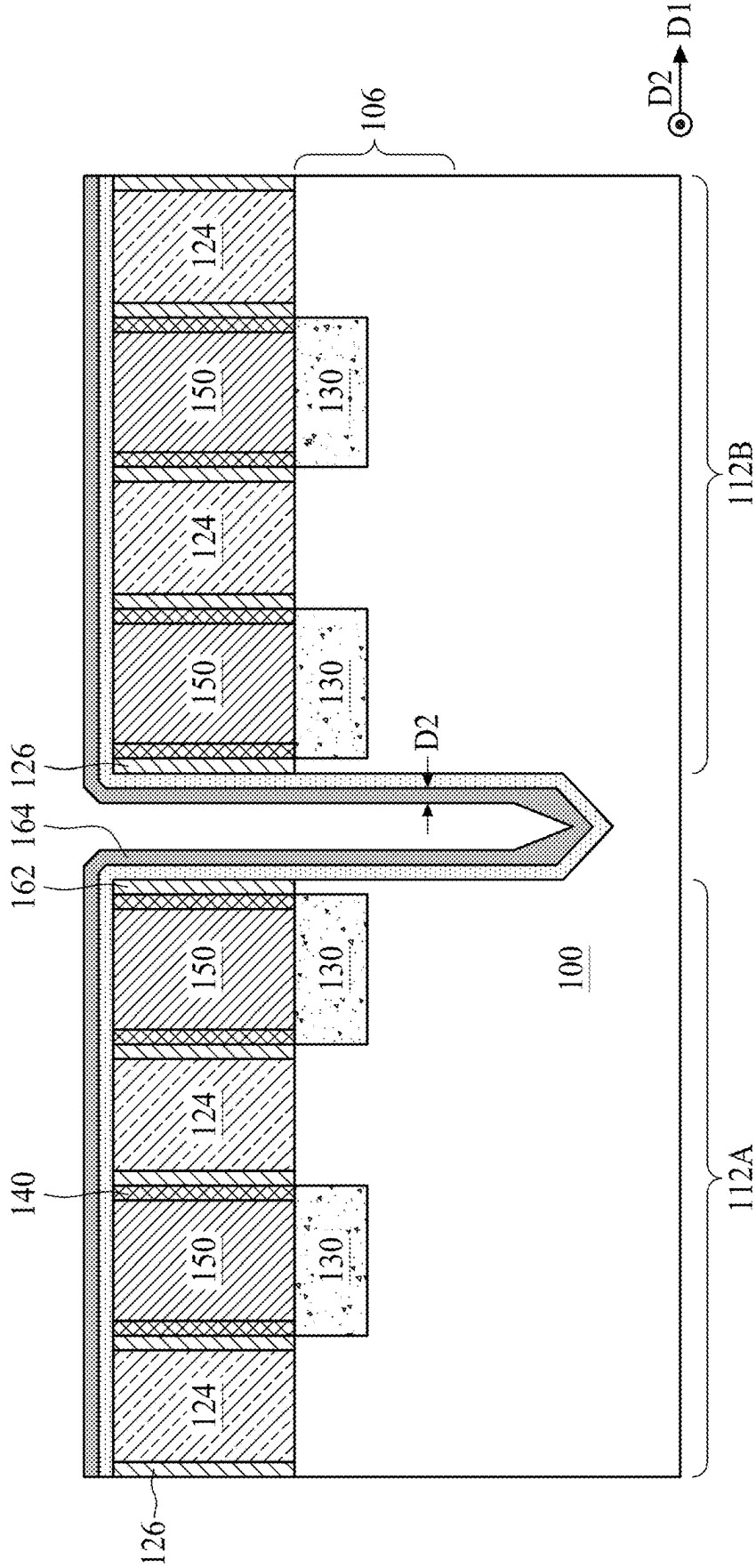


FIG. 13B

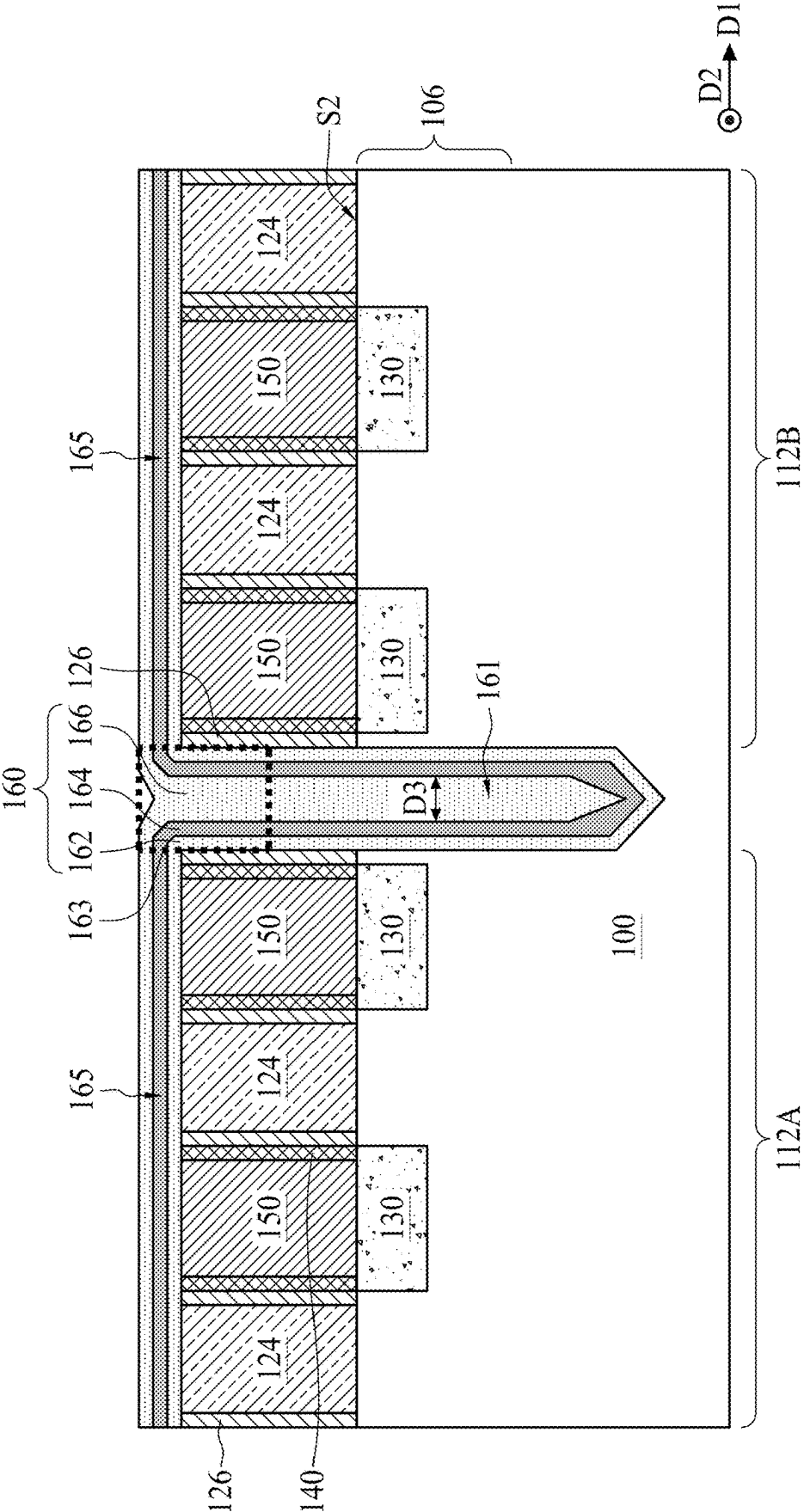


FIG. 13C

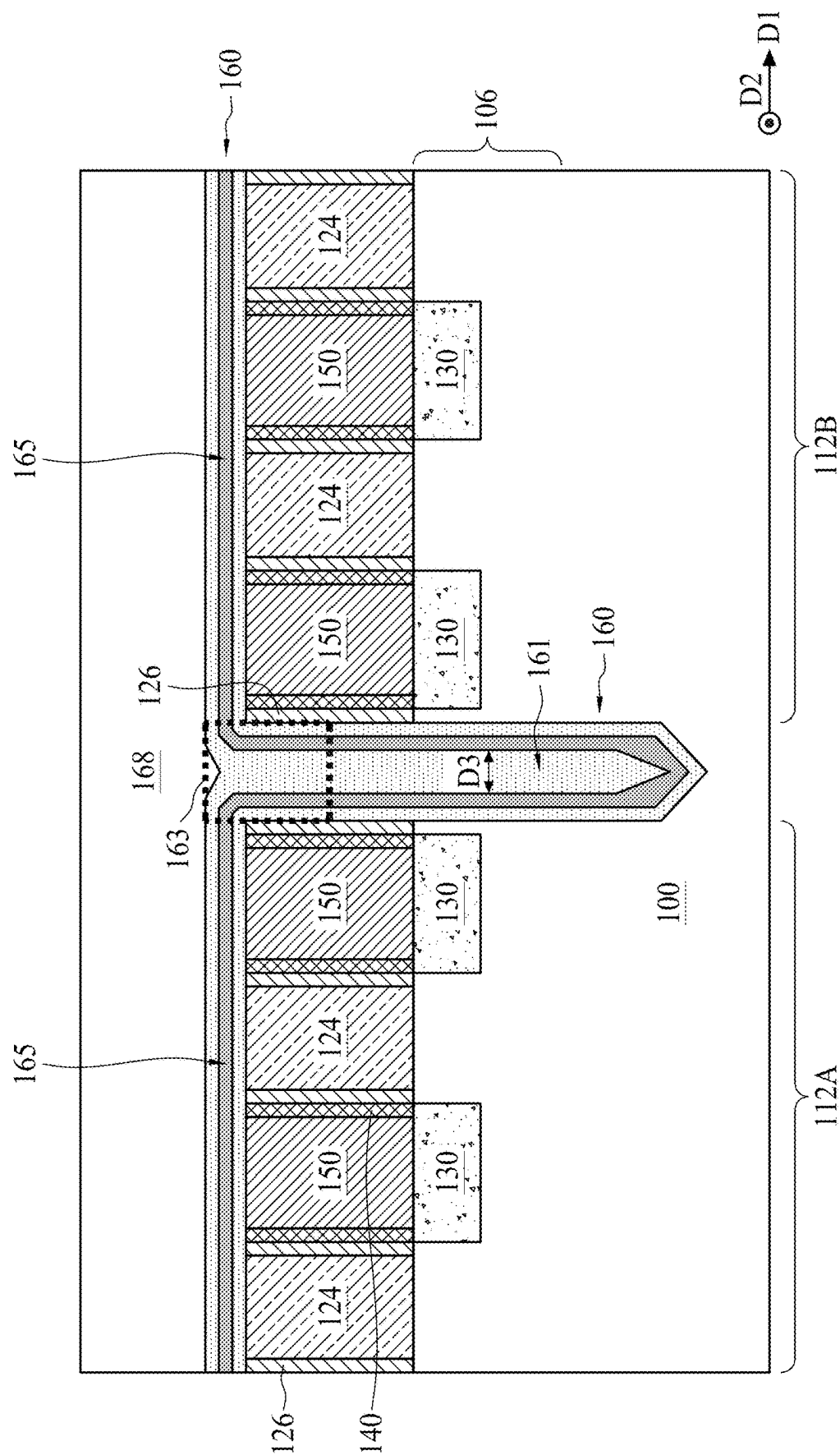
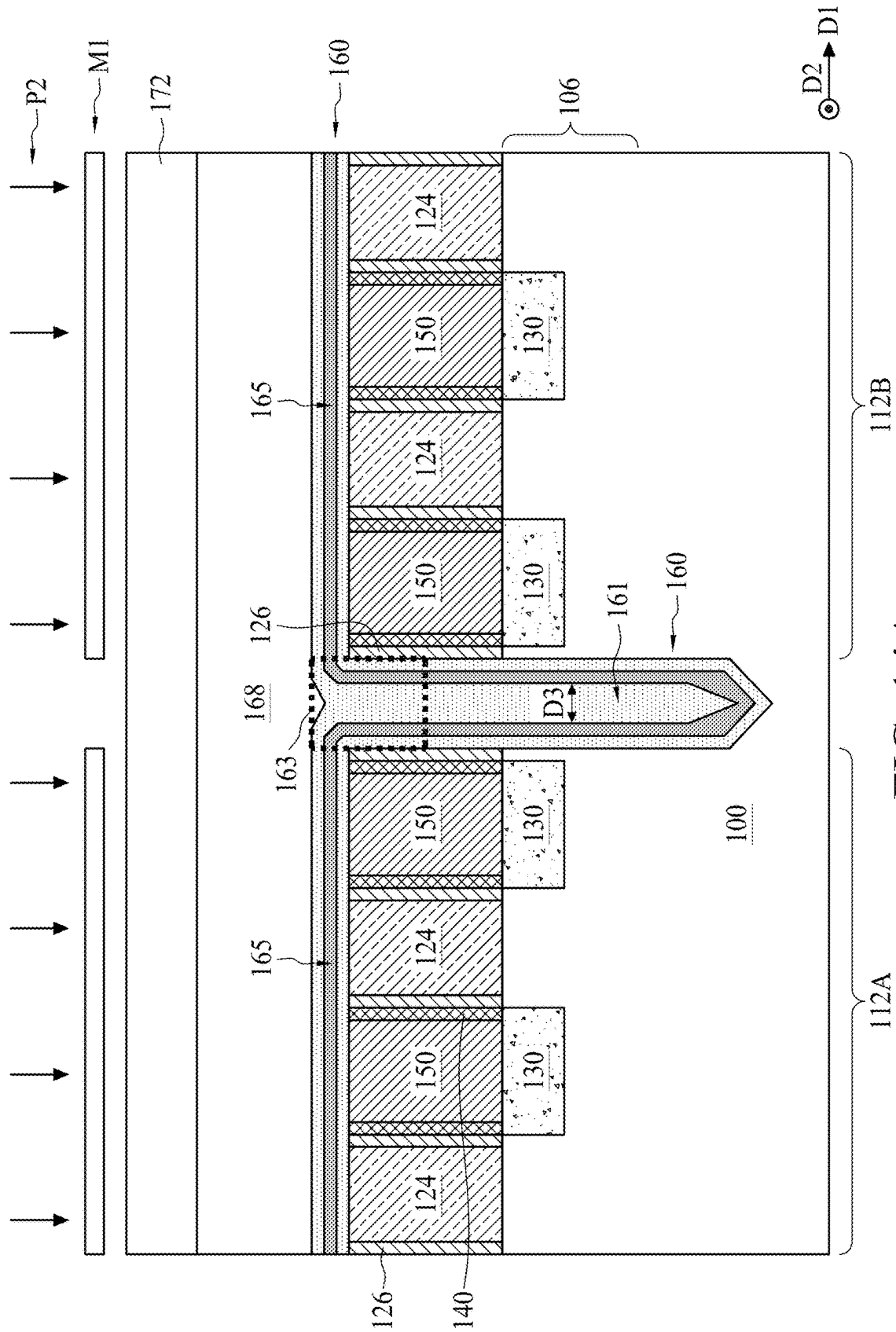
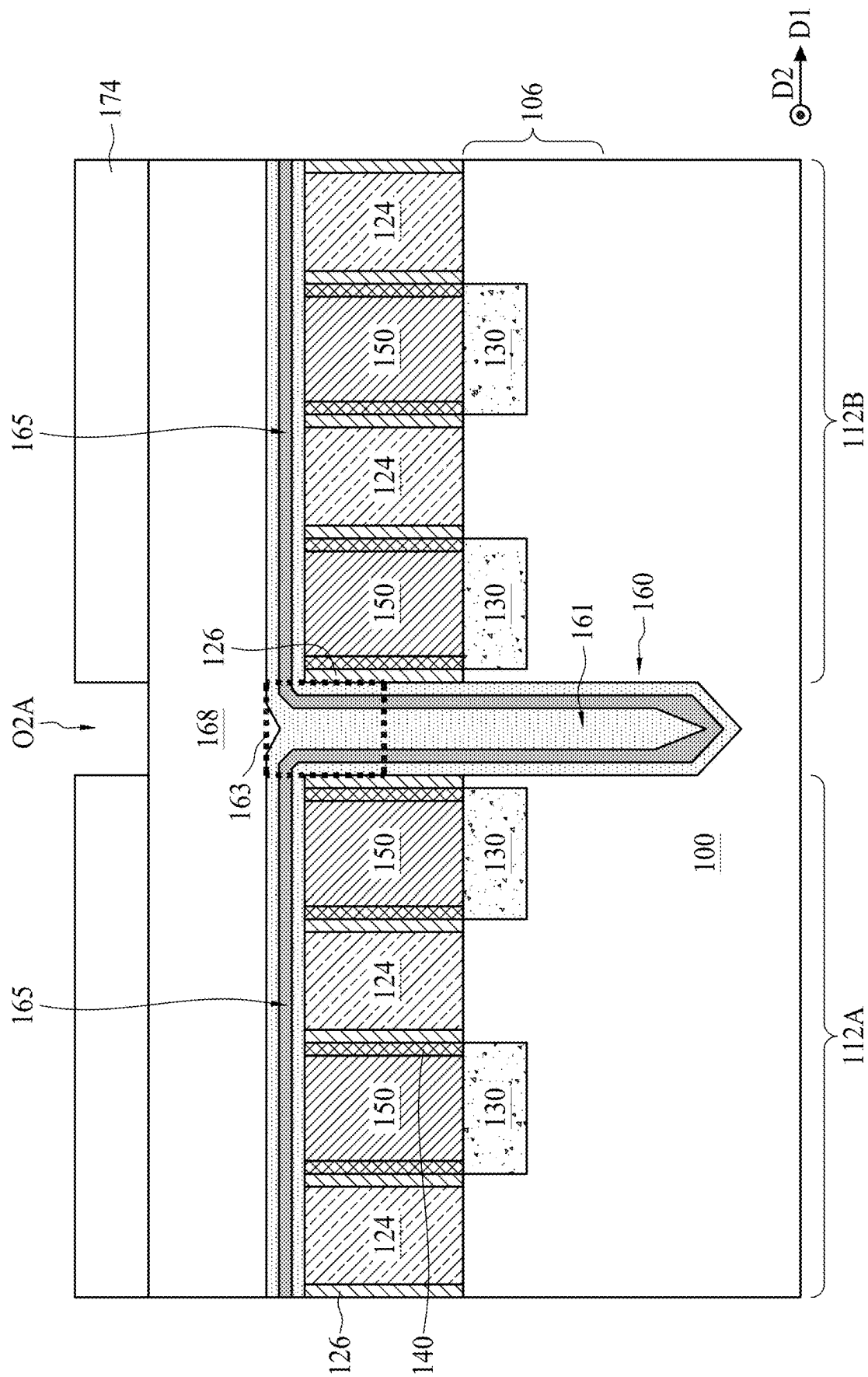


FIG. 13D





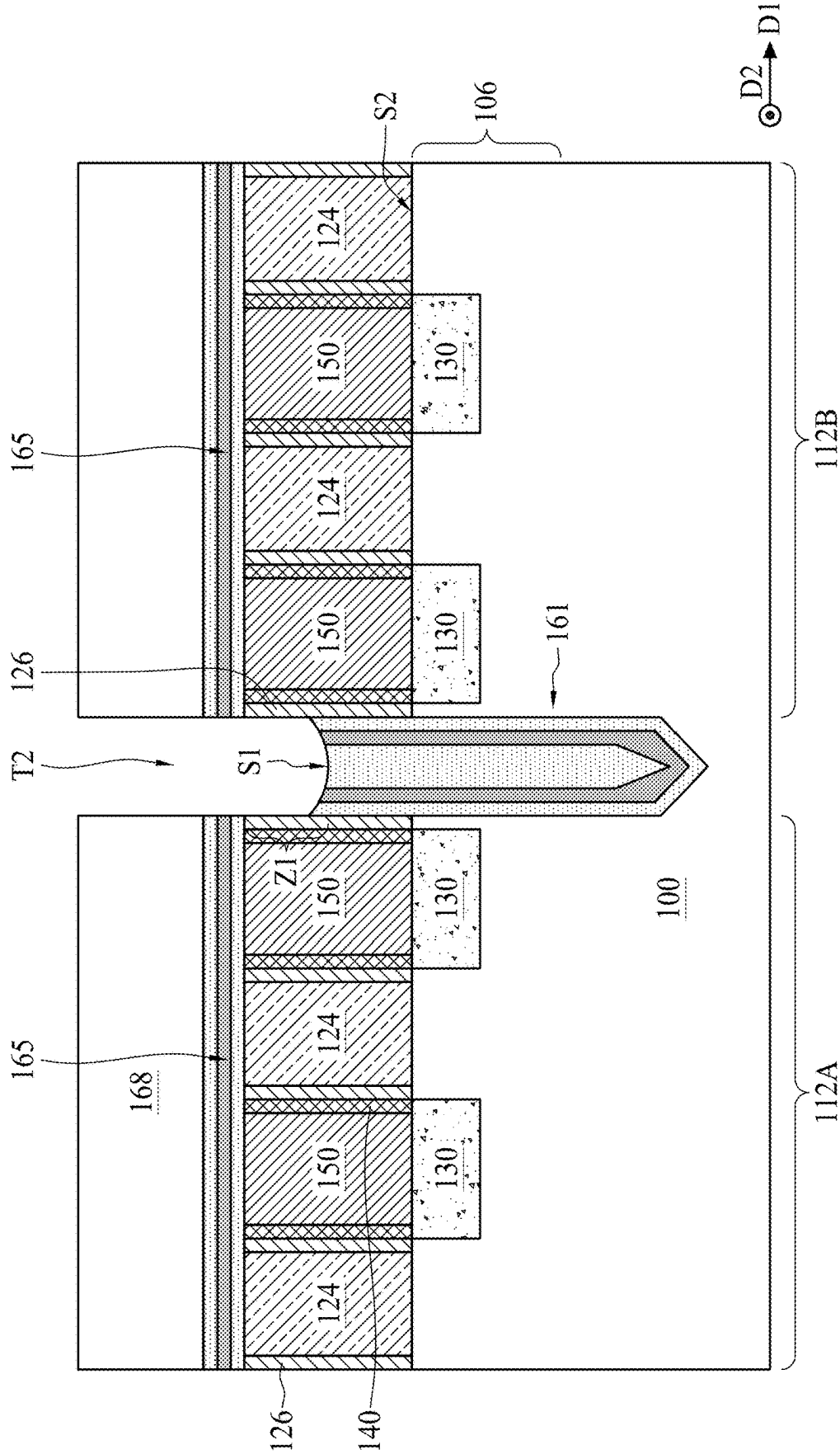


FIG. 15A

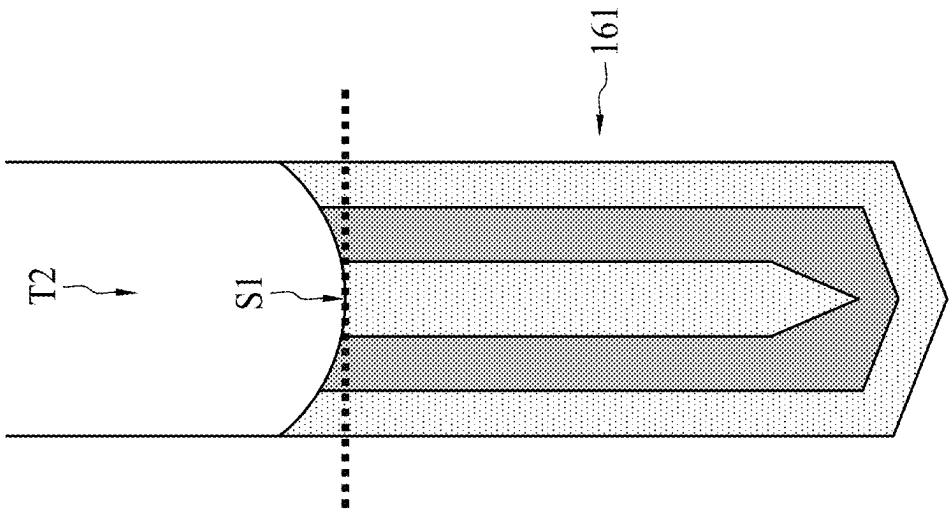


FIG. 15B

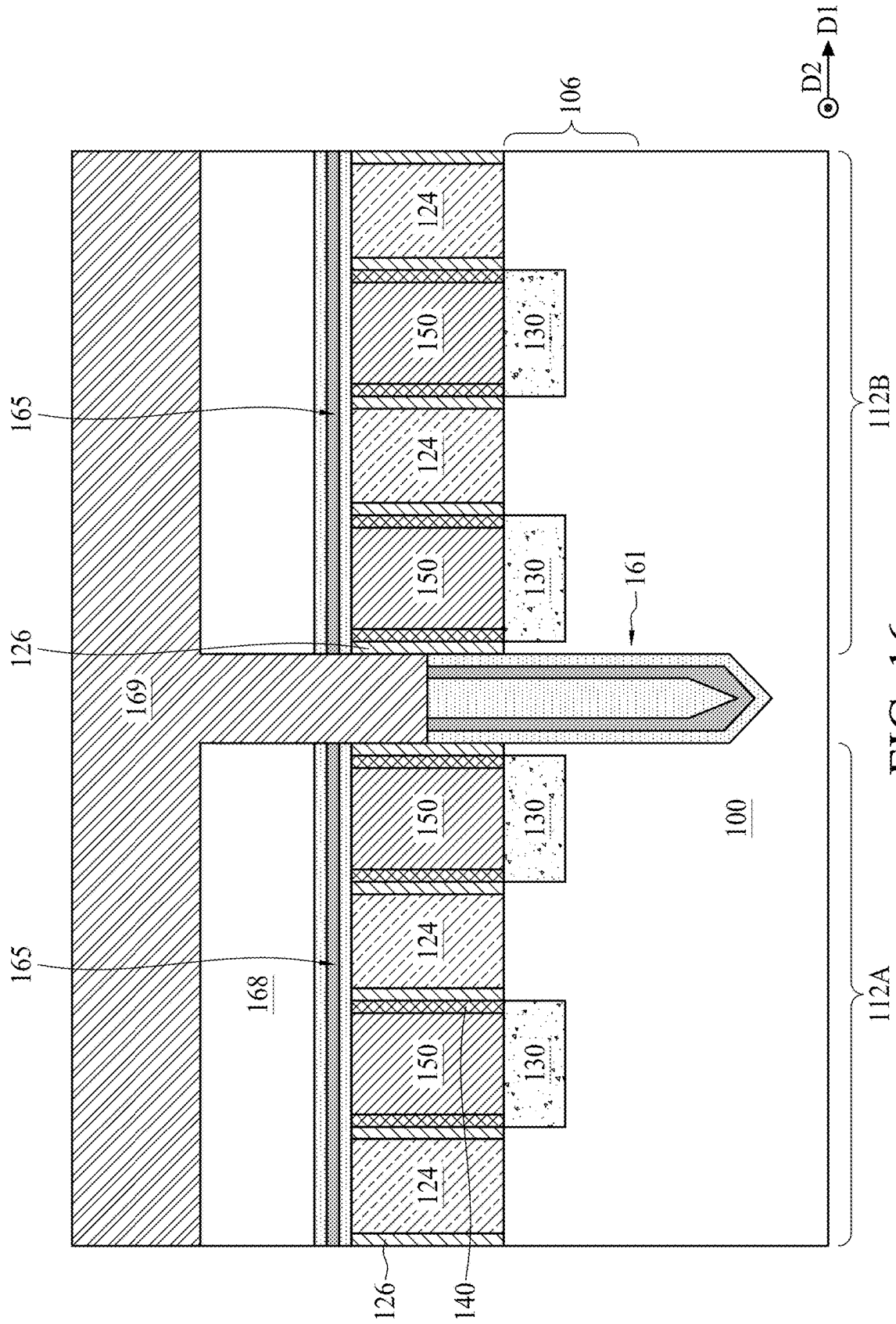


FIG. 16

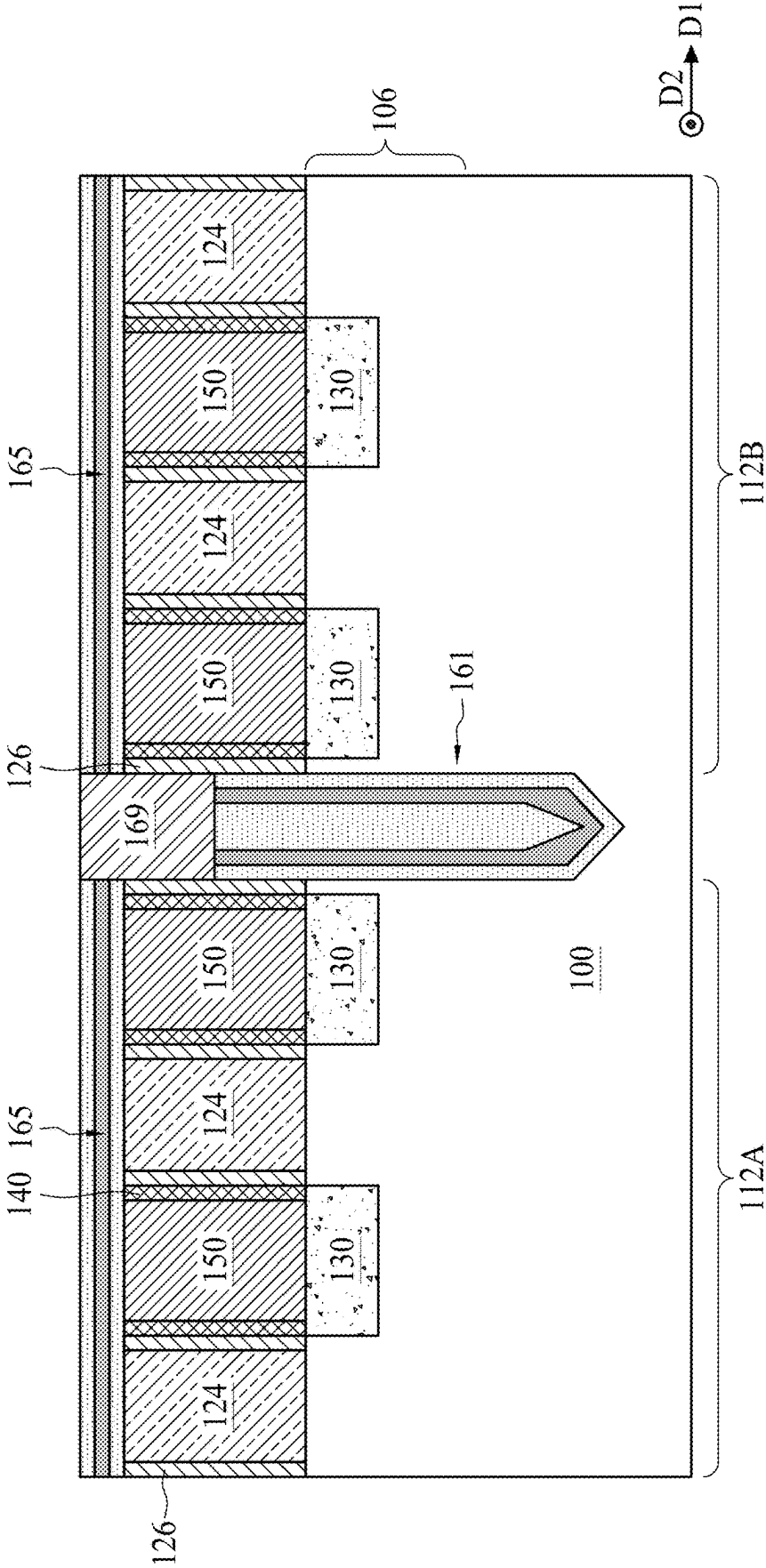


FIG. 17A

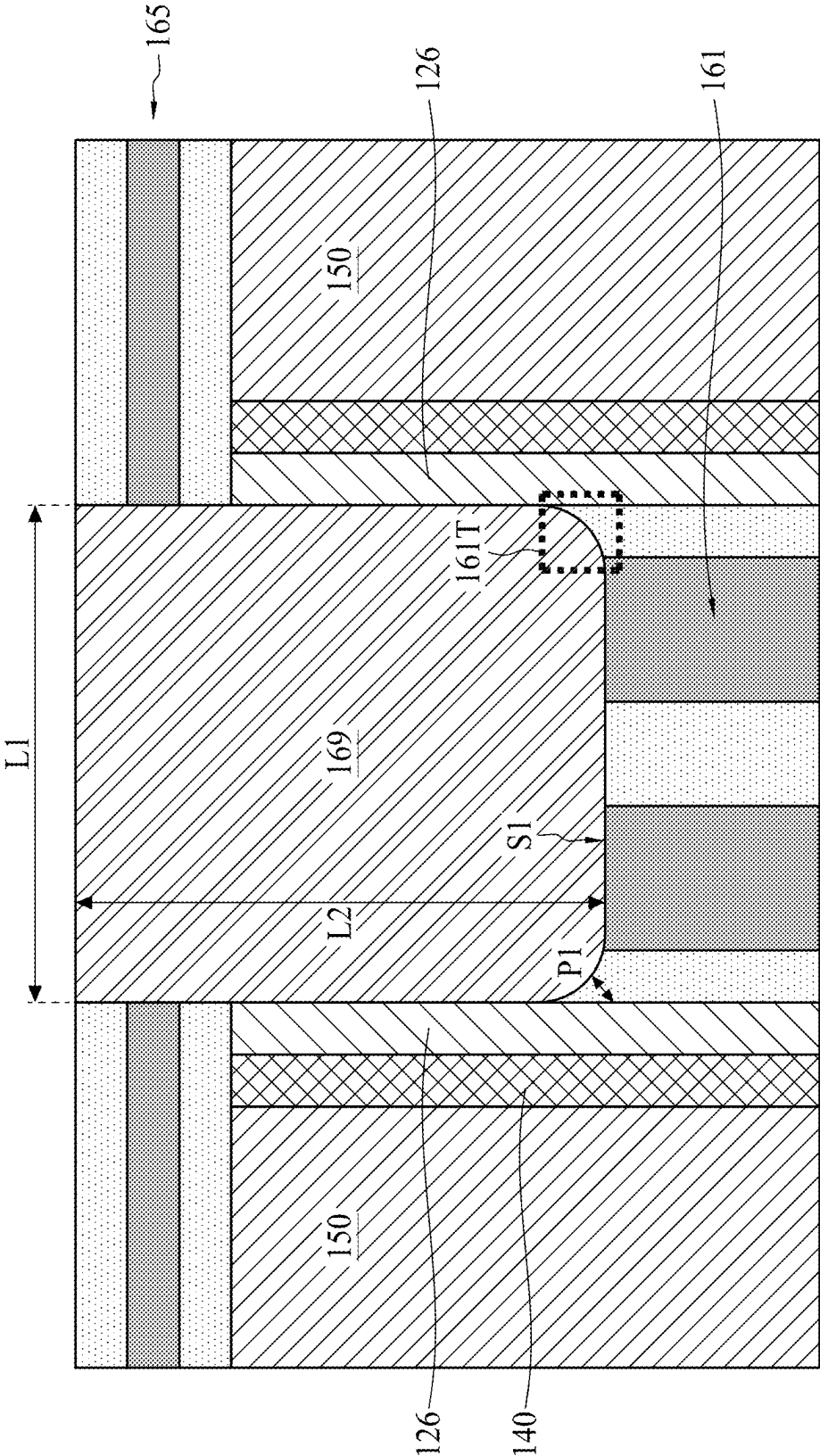
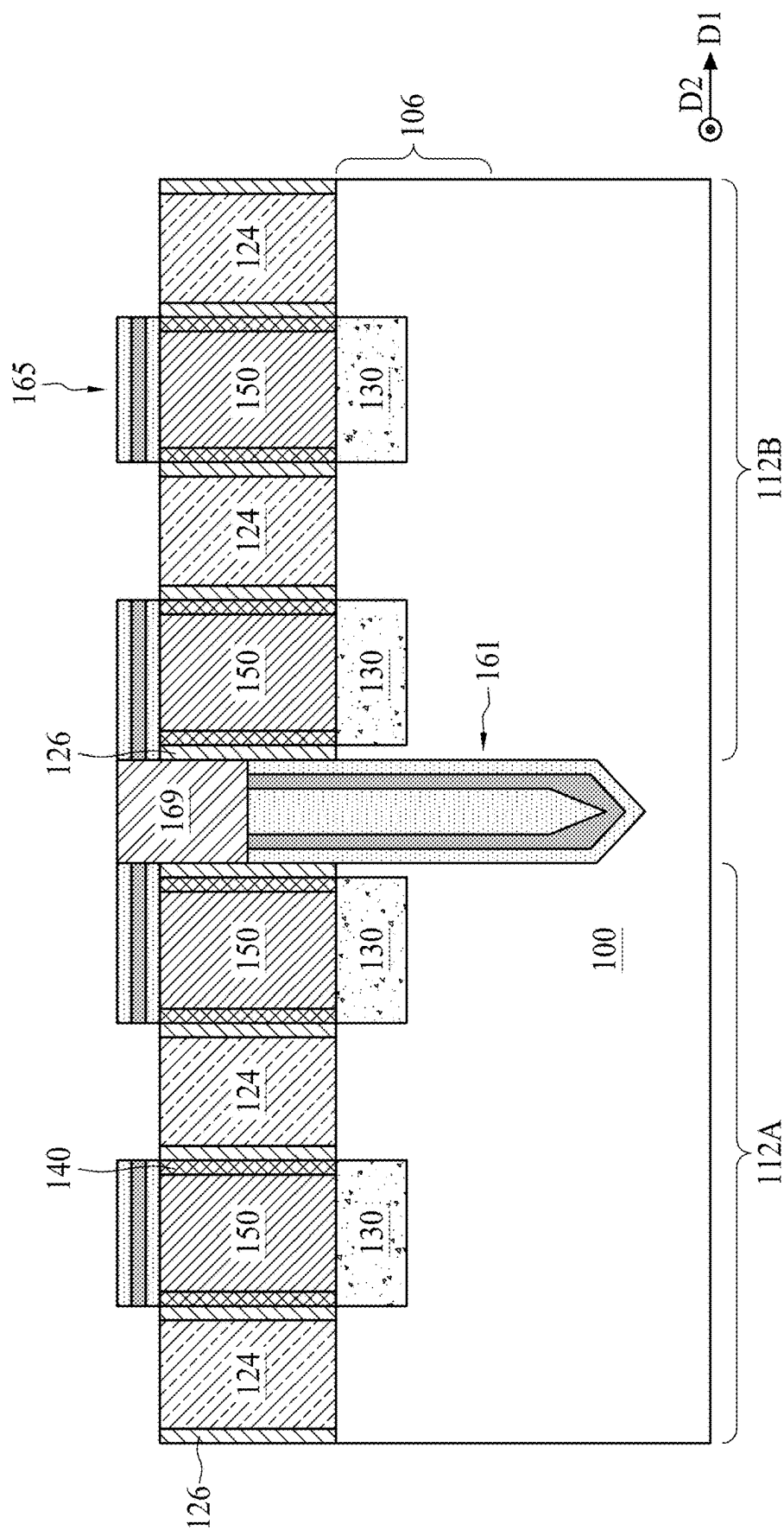


FIG. 17B



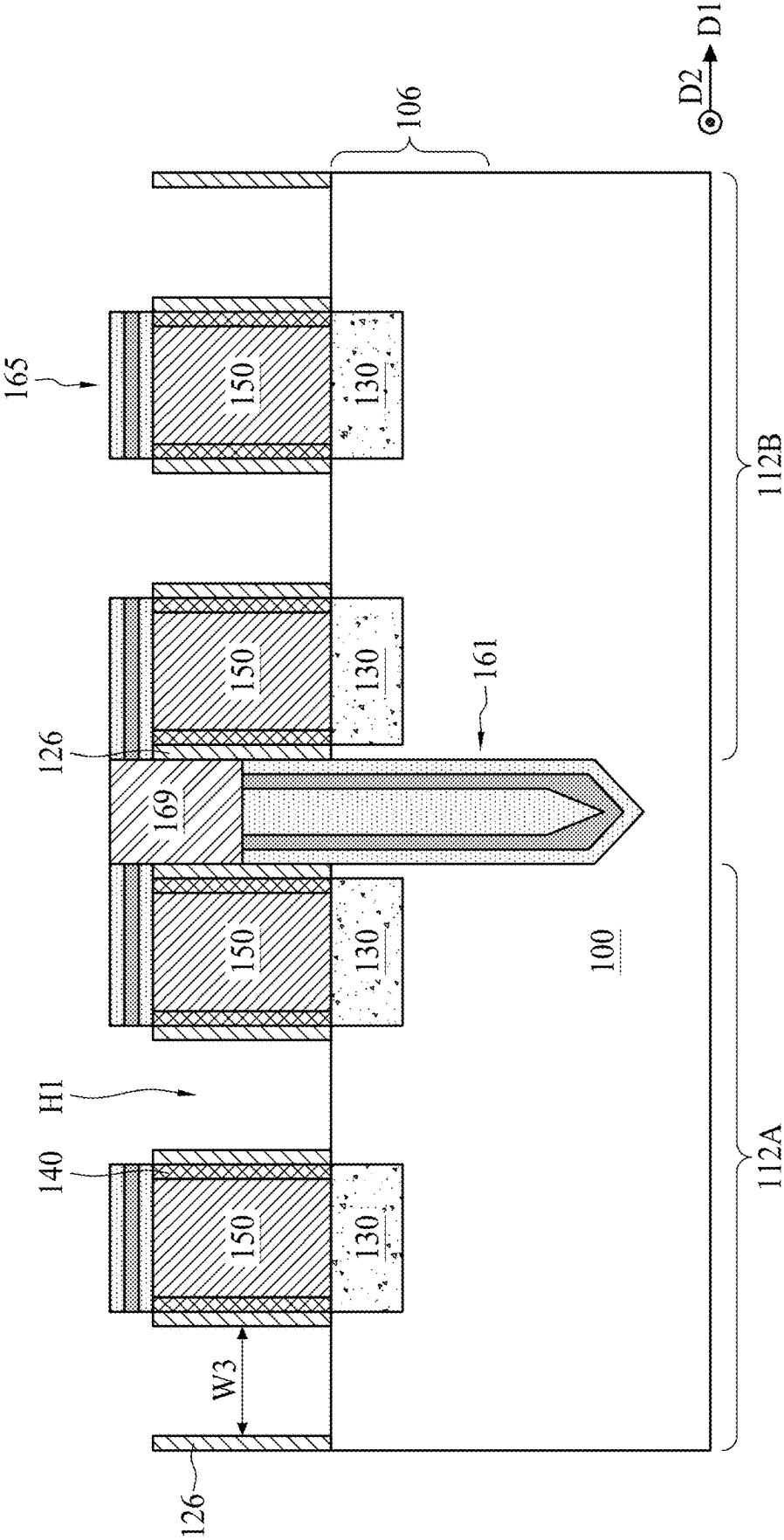
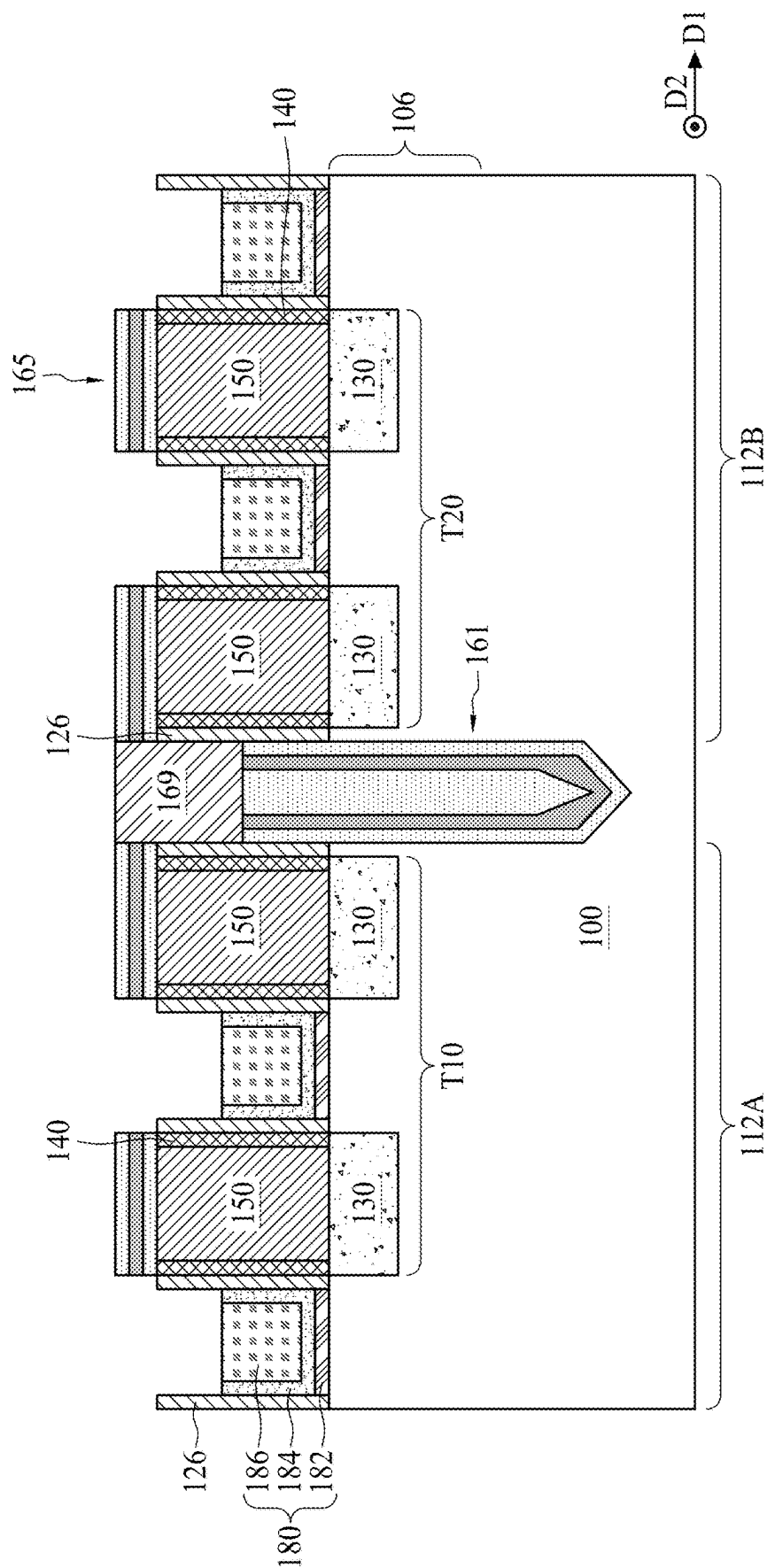
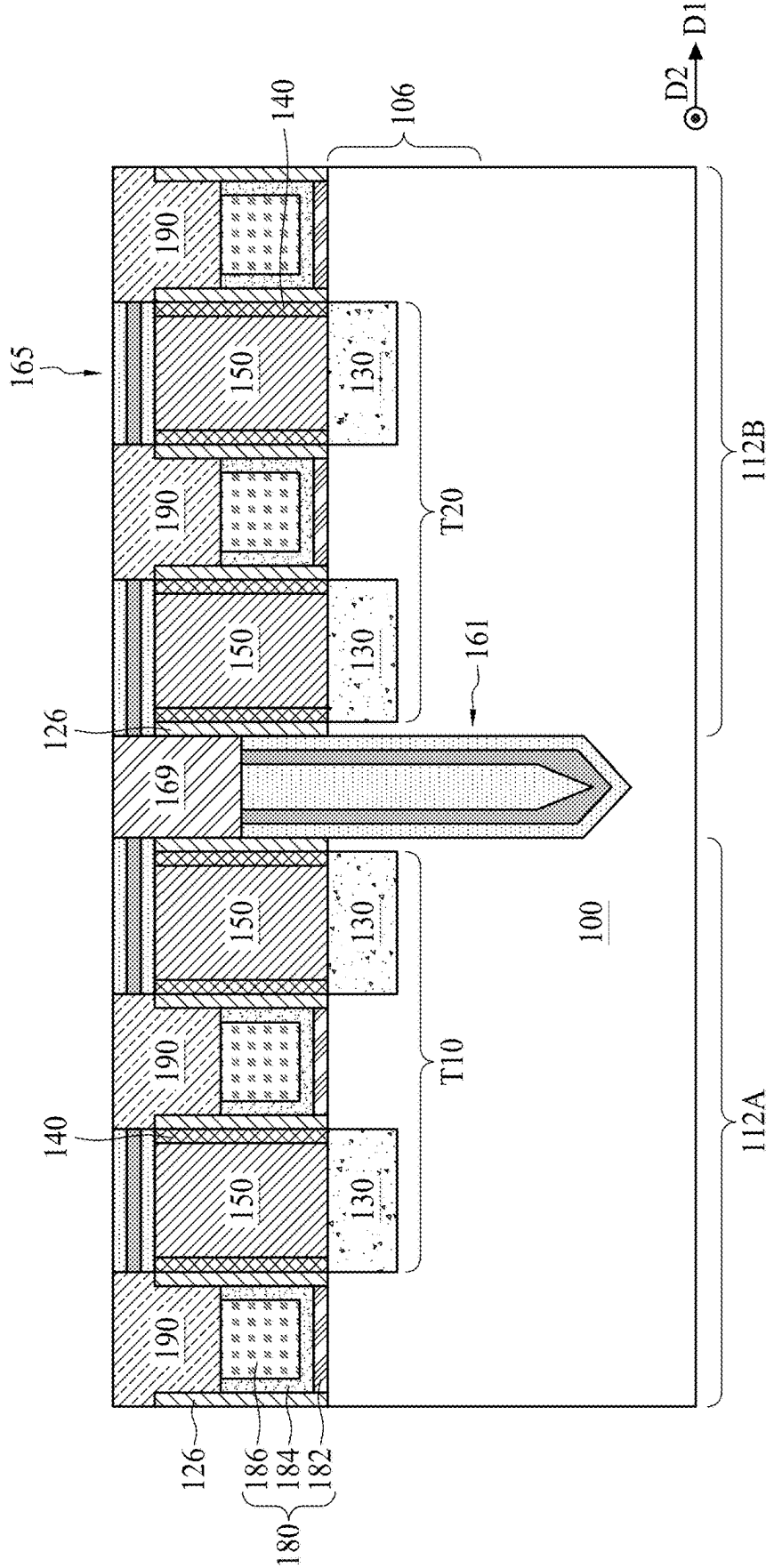


FIG. 18B





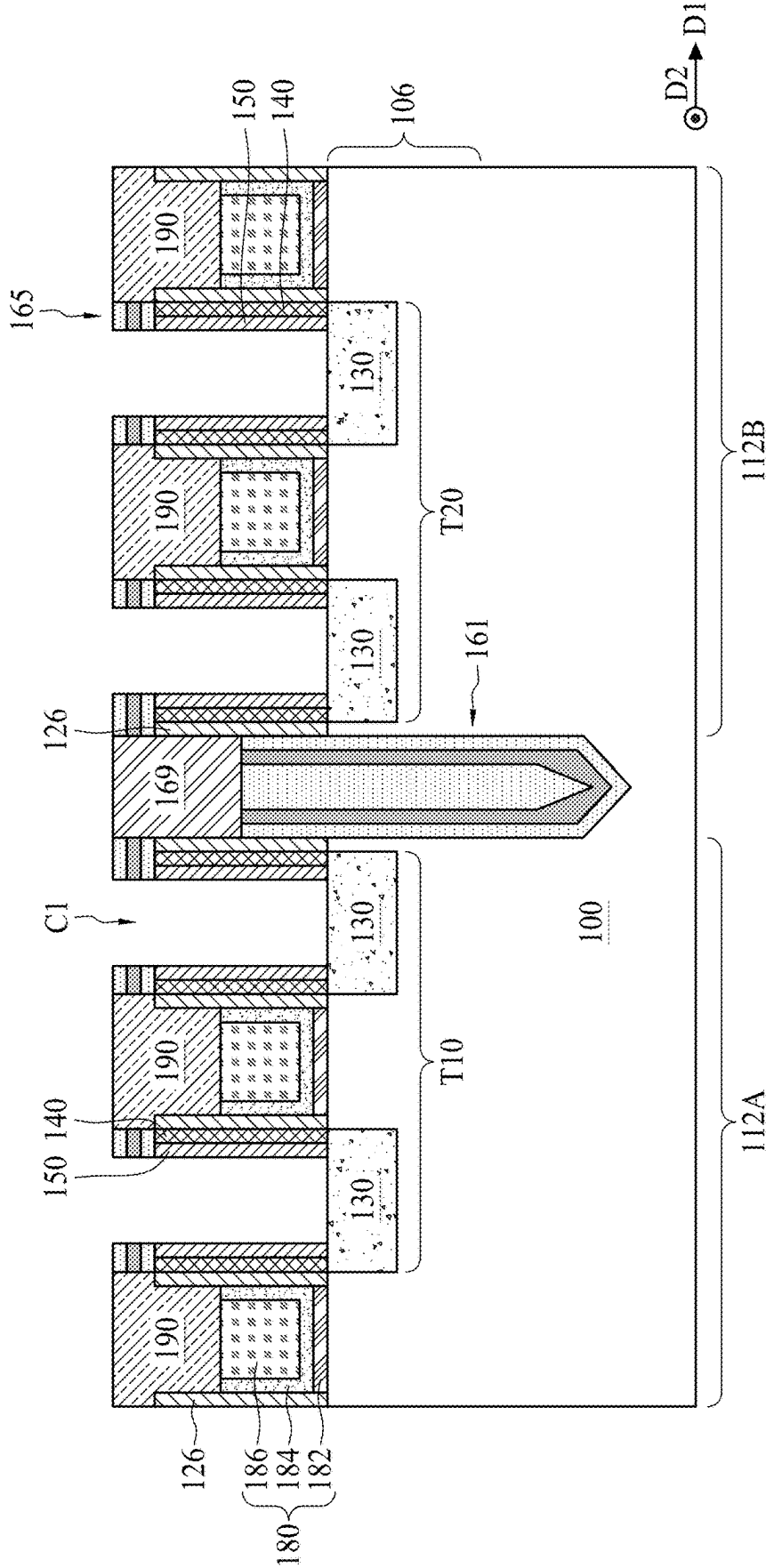


FIG. 21

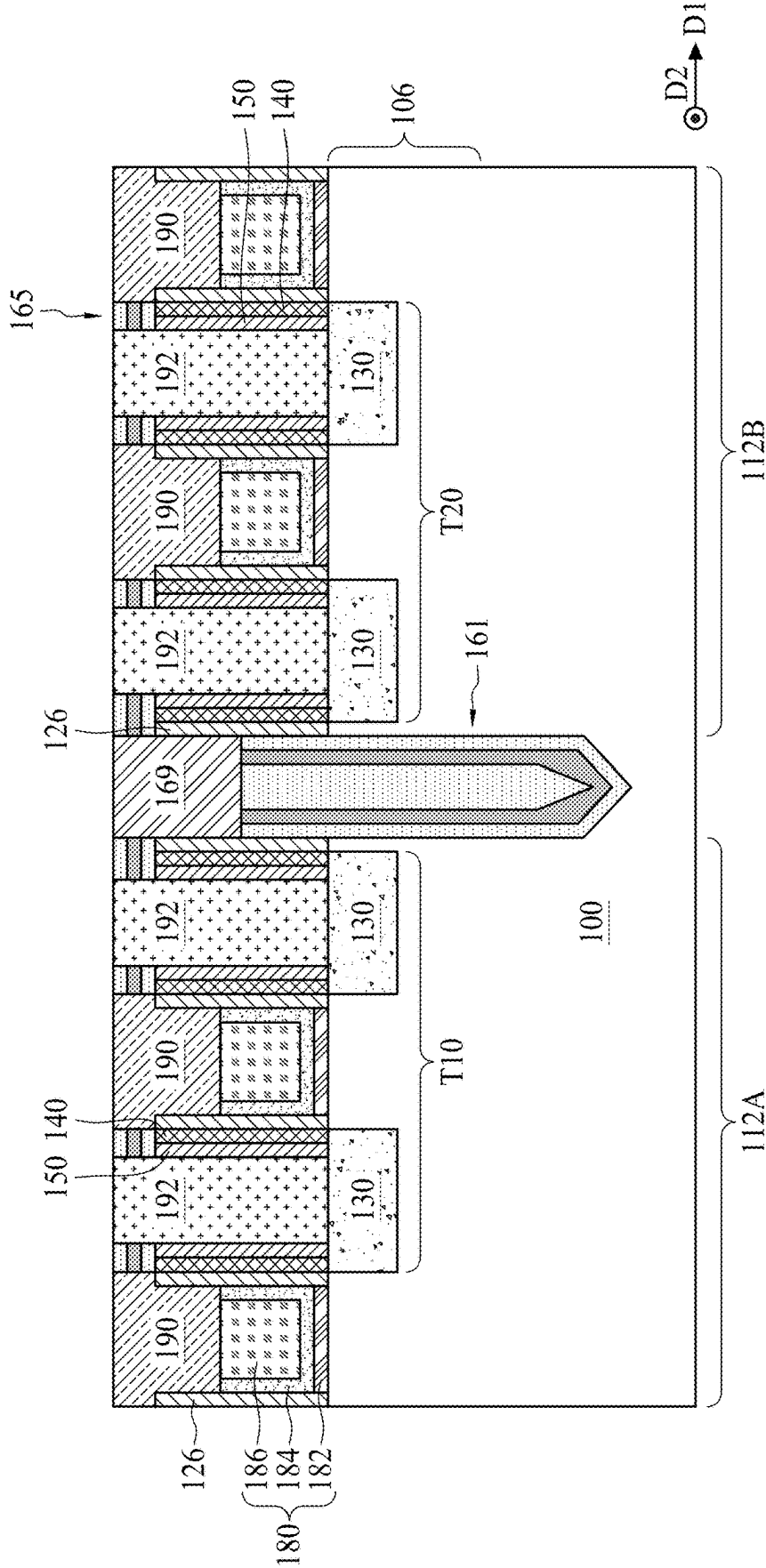


FIG. 22

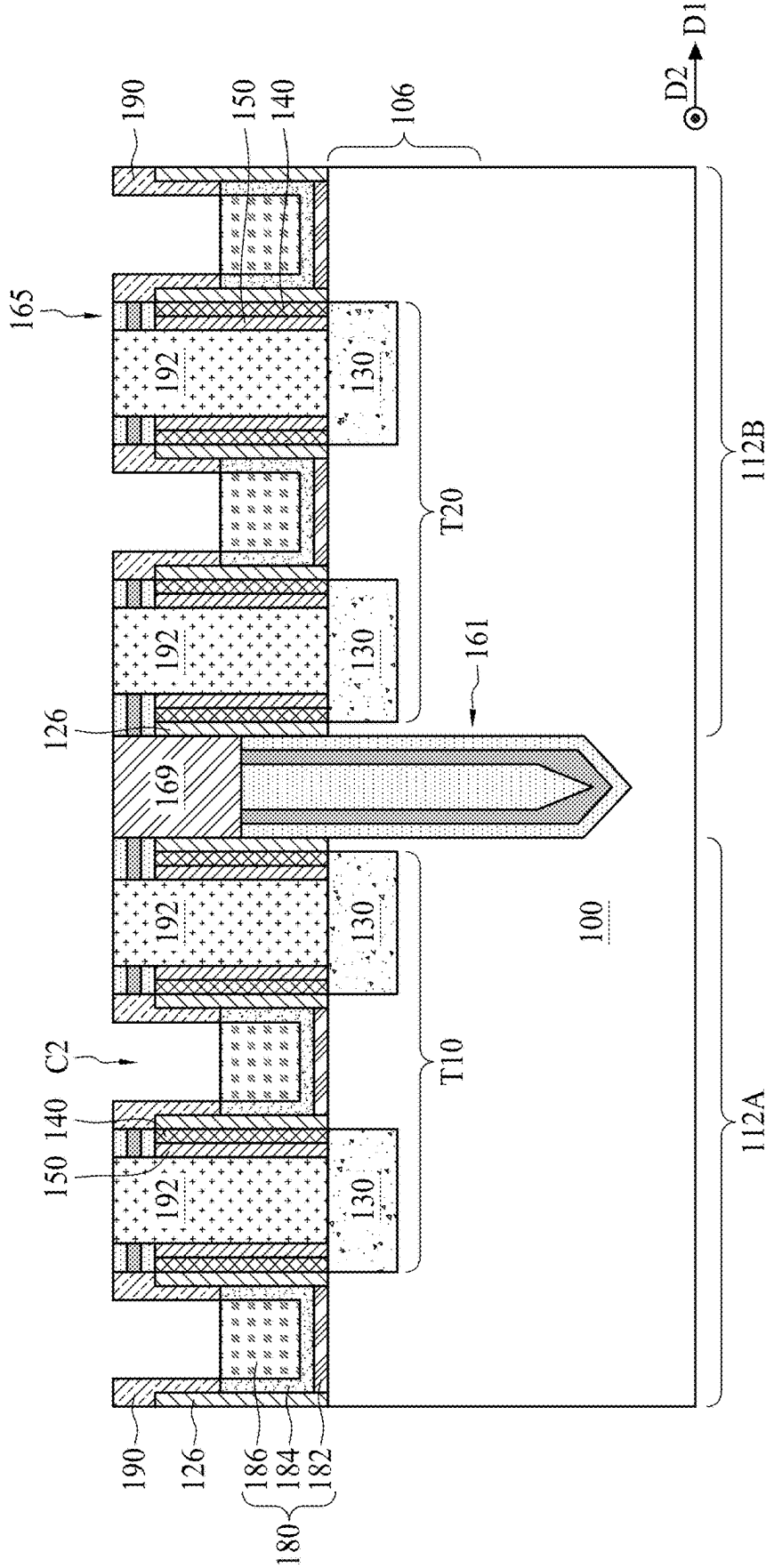
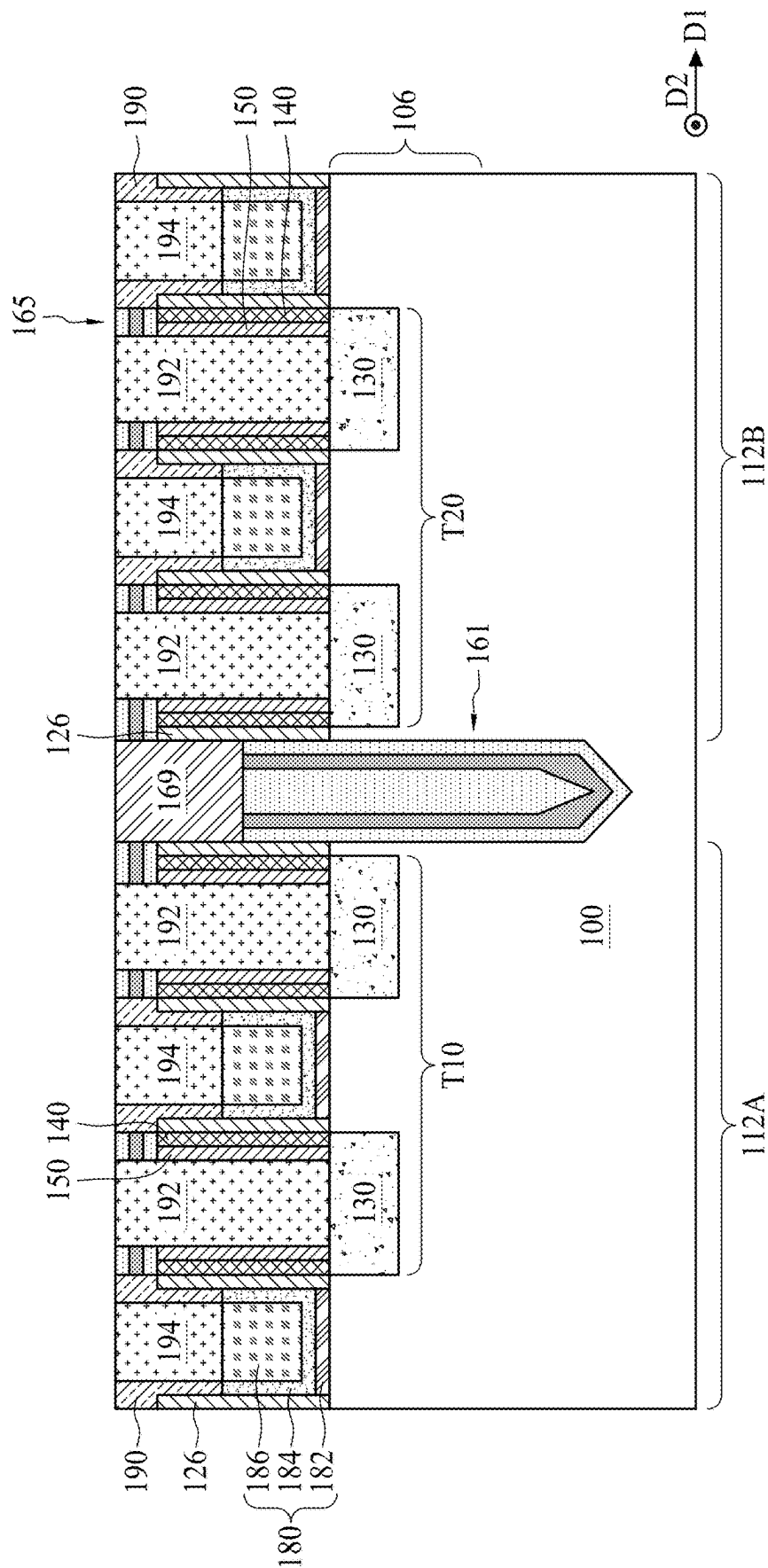


FIG. 23



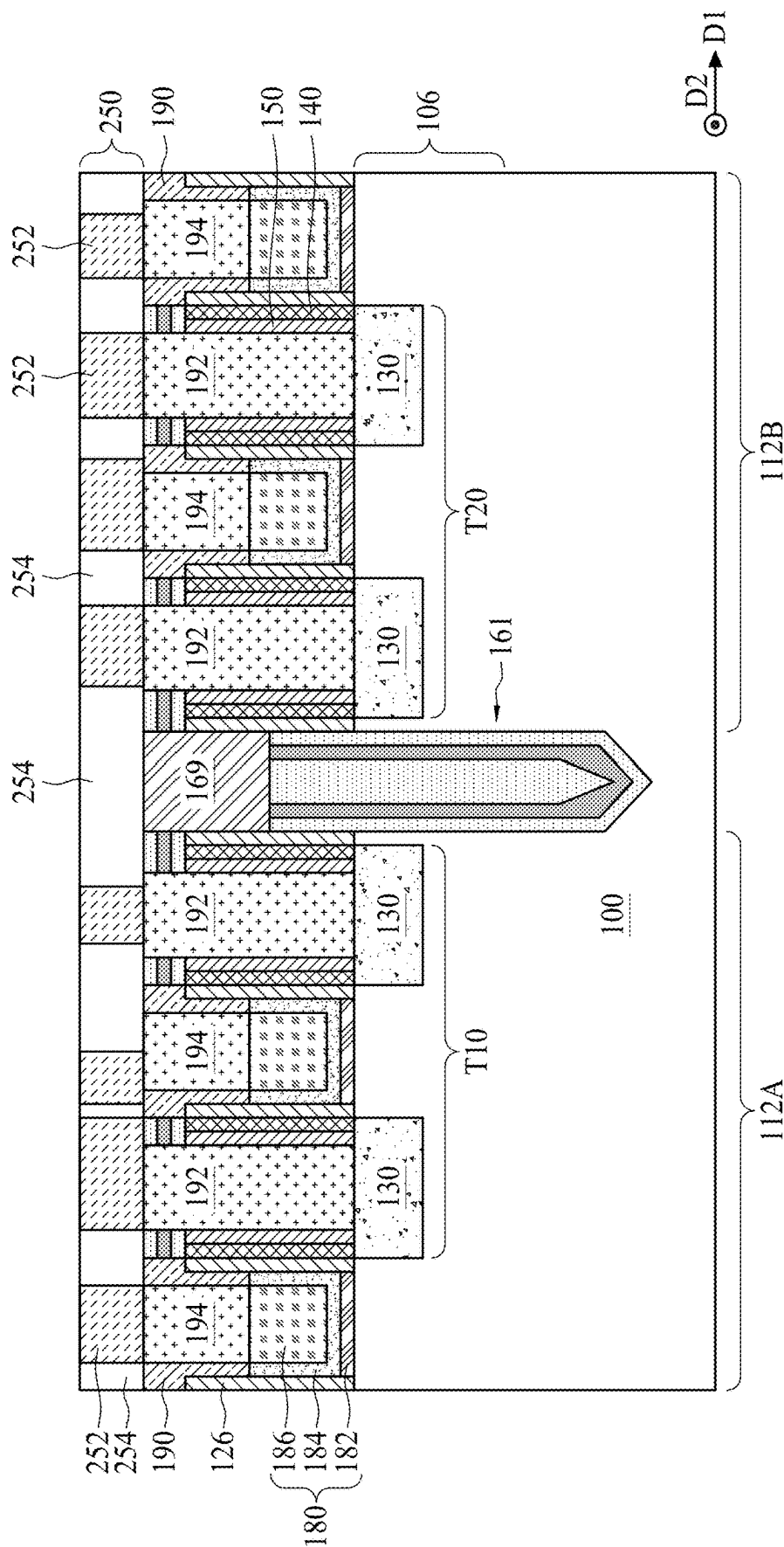
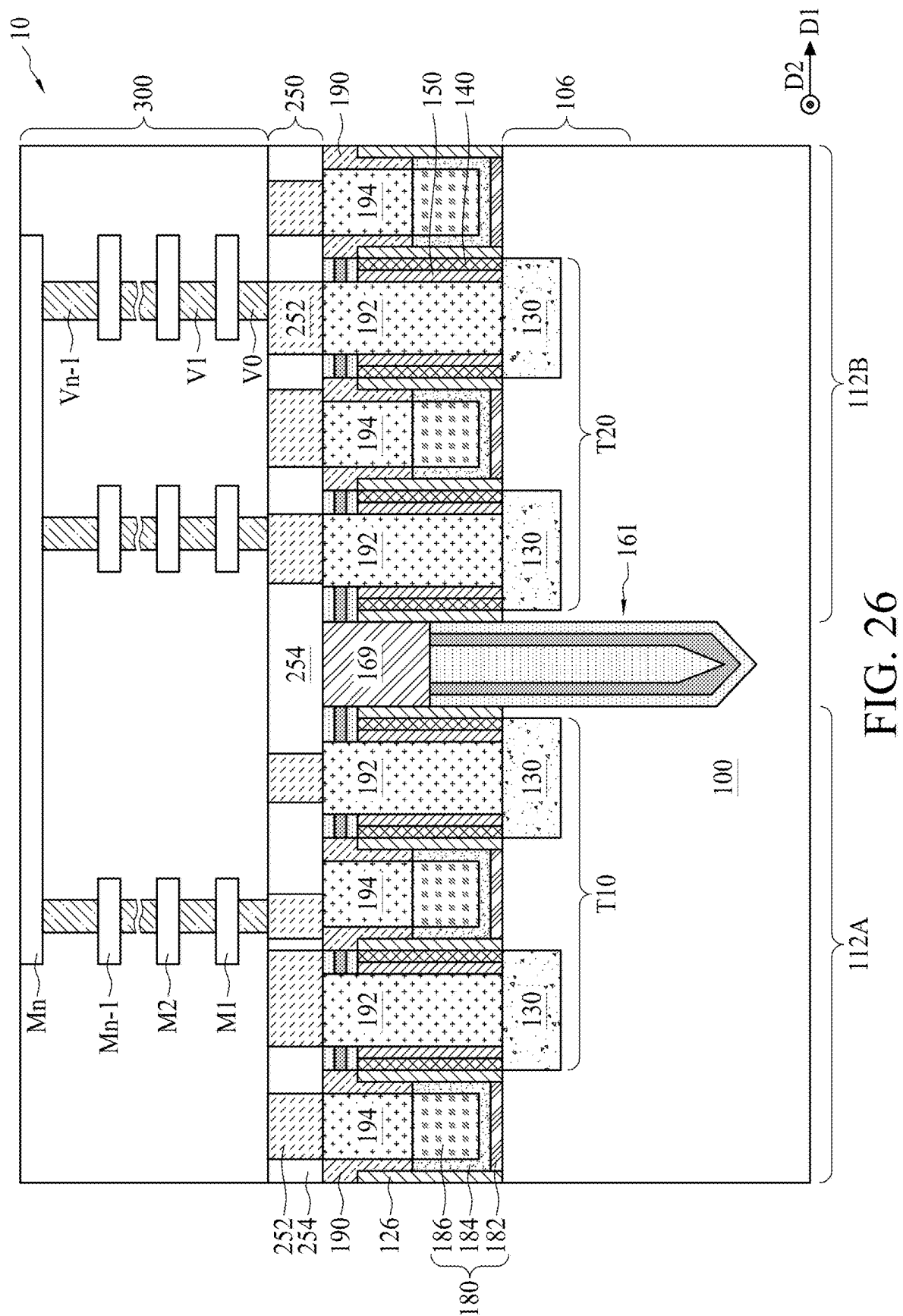


FIG. 25



SEMICONDUCTOR STRUCTURE INCLUDING ISOLATION STRUCTURE WITH DIELECTRIC STACK AND METHOD OF FORMING THE SAME

BACKGROUND

Semiconductor structures are used in a variety of electronic applications, such as personal computers, cellular telephones, digital cameras, and other electronic equipment. Semiconductor structures are typically fabricated by sequentially depositing insulating or dielectric layers, conductive layers, and semiconductive layers of material over a semiconductive substrate, and patterning the various material layers using lithography to form circuit components and elements thereon.

The semiconductor industry continues to increase the integration density of various electronic components (e.g., transistors, diodes, resistors, capacitors, etc.) by achieving continual reductions in minimum feature size, which allow more components to be integrated into a given area.

BRIEF DESCRIPTION OF THE DRAWINGS

Aspects of the embodiments of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various structures are not drawn to scale. In fact, the dimensions of the various structures may be arbitrarily increased or reduced for clarity of discussion.

FIG. 1 is a schematic cross-sectional view of a semiconductor structure, in accordance with some embodiments of the present disclosure.

FIG. 2 is a flow diagram showing a method of fabricating a semiconductor structure, in accordance with some embodiments of the present disclosure.

FIGS. 3, 4A to 4D, 5A, 6 to 10, 11A, 11B, 12A, 13A to 13D, 14A, 14B, 15A, 16, 17A, 18A, 18B, and 19 to 26 are schematic cross-sectional views and plan views illustrating sequential operations of the method shown in FIG. 2, in accordance with some embodiments of the present disclosure.

FIGS. 4E, 5B and 12B are schematic plan views of FIGS. 4D, 5A and 12A, respectively, in accordance with some embodiments of the present disclosure.

FIGS. 15B and 17B are enlarged views of FIGS. 15A and 17A, respectively, in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION OF THE DISCLOSURE

The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In some embodiments, the present disclosure may repeat reference numerals and/or letters in the

various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

Notwithstanding that the numerical ranges and parameters setting forth the broad scope of the disclosure are approximations, the numerical values set forth in the specific examples are reported as precisely as possible. Any numerical value, however, inherently contains certain errors necessarily resulting from the standard deviation found in the respective testing measurements. Also, as used herein, the terms “substantially,” “approximately,” or “about” generally mean within a value or range which can be contemplated by people having ordinary skill in the art. Alternatively, the terms “substantially,” “approximately,” or “about” mean within an acceptable standard error of the mean when considered by one of ordinary skill in the art. People having ordinary skill in the art can understand that the acceptable standard error may vary according to different technologies. Other than in the operating/working examples, or unless otherwise expressly specified, all of the numerical ranges, amounts, values and percentages, such as those for quantities of materials, durations of times, temperatures, operating conditions, ratios of amounts, and the likes thereof disclosed herein, should be understood as modified in all instances by the terms “substantially,” “approximately,” or “about.” Accordingly, unless indicated to the contrary, the numerical parameters set forth in the present disclosure and attached claims are approximations that can vary as desired. At the very least, each numerical parameter should at least be construed in light of the number of reported significant digits and by applying ordinary rounding techniques. Ranges can be expressed herein as from one endpoint to another endpoint or between two endpoints. All ranges disclosed herein are inclusive of the endpoints, unless specified otherwise.

A metal-oxide-semiconductor field-effect-transistor (MOSFET) device is a device including a source terminal and a drain terminal separated by a region, referred to as a channel region of the MOSFET device. The MOSFET device is modulated by applying a voltage to an electrode (gate) which overlies the channel region and is separated from it by a thin insulating layer (generally referred to as a gate dielectric). The simple nature of the MOSFET devices and their minimal heat dissipation permit an extraordinary degree of miniaturization and consequently a high density of circuits.

A “continuous poly on oxide definition edge” or a “continuous poly on oxide diffusion edge” (CPODE) structure refers to an isolation structure that separates multiple active regions (i.e., oxide definition regions) by transforming a dummy structure that crosses the active regions into the isolation structure. The CPODE structure may be arranged proximal to a boundary between two active regions for achieving a desired electrical isolation performance. In the present disclosure, a CPODE structure is formed in the front-end-of-line (FEOL) stage of fabricating a semiconduc-

tor structure. The FEOL stage refers to processes of forming transistors and elements proximal to the transistors. The middle-end-of-line (MEOL) stage refers to processes of forming conductive contacts or vias electrically interconnecting the underlying transistors and overlying interconnection wires formed in the metallization layers, and the back-end-of-line (BEOL) stage refers to processes of forming interconnection or metallization layers over the transistors for routing the features in the transistors. The processes in the FEOL and MEOL stages provide a smaller critical dimension (CD) between elements, as compared to the processes in the BEOL stage.

FIG. 1 is a schematic cross-sectional view of a semiconductor structure 10, in accordance with some embodiments of the present disclosure. The semiconductor structure 10 includes a substrate 100 and a fin structure 106 on the substrate 100. A CPODE structure 161 including a dielectric stack penetrates the fin structure 106 and extends into the substrate 100. The dielectric stack includes an oxide-nitride-oxide ("ONO") structure. A top surface of the CPODE structure 161 is higher than a top surface of the fin structure 106 by about 30 Å to about 60 Å. The CPODE structure 161 separates to the fin structure 106 into a first active region 112A and a second active region 112B.

A first transistor T10 and a second transistor T20 adjacent to the first transistor T10 are disposed on the fin structure 106. Each of the first transistor T10 and the second transistor T20 includes a gate structure 180 and a pair of epitaxial features 130 on opposite sides of the gate structure 180. The gate structure 180 functions as a gate terminal of the transistor T10 or T20, and the pair of epitaxial features 130 adjacent to the gate structure 180 functions as source/drain terminals of the transistor T10 or T20.

A gate spacer 126 is disposed on a sidewall of the gate structure 180. The gate structure 180 includes a gate dielectric layer 182, a high-k material layer 184 and a gate electrode 186. In some embodiments, a height of the gate structure 180 is substantially lower than a height of the gate spacer 126. The gate dielectric layer 182 is a planar film disposed on the fin structure 106. The high-k material layer 184 may be a planar or U-shaped layer disposed on the gate dielectric layer 182. The gate electrode 186 may be a planar or U-shaped layer surrounded by the high-k material layer 184. In some embodiments, the gate spacer 126 surrounds the gate dielectric layer 182, the high-k material layer 184 and the gate electrode 186. In some embodiments, a top surface of the gate electrode 186 is lower than a top surface of the gate spacer 126. A hard mask 190 is formed over the gate structure 180. A protective layer 169 is directly disposed on the dielectric stack of the CPODE structure 161. The first transistor T10 and the second transistor T20 are separated by the CPODE structure 161 and the protective layer 169. The top surface of the CPODE structure 161 is lower than a top surface of the gate structure 180 or the gate spacer 126. In some embodiments, top surfaces of the hard mask 190 and the protective layer 169 are substantially coplanar.

A conductive via 192 is disposed on and electrically coupled to the underlying epitaxial feature 130. A conductive via 194 is disposed on and electrically coupled to the underlying gate electrode 186. An interconnect layer 250 is disposed over the conductive vias 192 and 194. The interconnect layer 250 includes a metal line layer 252 embedded in an interlayer dielectric (ILD) layer 254. The metal line layer 252 is disposed over and electrically coupled to the conductive vias 192 and 194. The metal line layer 252 may

be electrically coupled to the transistor T10 or the transistor T20 through the conductive vias 192 and 194, respectively.

A metallization layer 300 is disposed over the interconnect layer 250. The metallization layer 300 includes a stack of metal line layers (wires) $M_1, \dots, M_{n-1}, M_n$ (the integer n denotes the layer index) disposed over one another and a stack of metal via layers $V_0, \dots, V_{n-1}$ (the integer n denotes the layer index) disposed over one another. A metal via layer V_n is interposed between an underlying metal line layer M_n and an overlying metal line layer M_{n+1} and electrically connects the metal line layer M_n and M_{n+1} . The number of n may be configured according to the design requirements of the metallization layer 300.

FIG. 2 is a flow diagram showing a method 200 of fabricating a semiconductor structure, in accordance with some embodiments of the present disclosure. FIGS. 3 to 26 are schematic cross-sectional views and plan views illustrating sequential operations of the method 200 shown in FIG. 2, in accordance with some embodiments of the present disclosure.

Referring to operation 201 of FIG. 2, a substrate 100 is provided, as shown in FIG. 2. In some embodiments, the substrate 100 includes a bulk material, such as bulk silicon. The substrate 100 may be a silicon-on-insulator (SOI) substrate. In some embodiments, the substrate 100 includes a suitable elementary semiconductor, such as germanium (Ge) or diamond. In some other embodiments, the substrate 100 includes a compound semiconductor, such as silicon germanium (SiGe), silicon carbide (SiC), gallium arsenide (GaAs), gallium nitride (GaN), indium phosphide (InP) or the like.

Referring to operation 203 of FIG. 2, a fin structure 106 is formed on the substrate 100, as shown in FIGS. 4A to 4E. A top portion of the substrate 100 may be patterned using one or more lithographic and etching operations. For example, a double-patterning or multi-patterning technique known in the art can be used to form the fin structure 106 on the substrate 100. Referring to FIG. 4A, a patterned mask 102 may be formed on the substrate 100. The patterned mask 102 may be a patterned photoresist in strips and include multiple openings exposing the underlying substrate 100. The patterned mask 102 is used as an etching mask during a subsequent etching operation. Referring to FIG. 4B, the substrate 100 is etched through the openings of the patterned mask 102 until multiple trenches 104 with a predetermined depth are formed. The trenches 104 may also be strips that are parallel to each other. Referring to FIG. 4C, after the patterned mask 102 is removed, multiple fin structures 106 are exposed. The fin structures 106 may be arranged in strips and protruded from a lower portion of the substrate 100. An ion implantation operation may be performed on the fin structures 106. For example, P-type dopants such as boron (B), gallium (Ga) and indium (In) ions, or N-type dopants such as phosphorus (P) and arsenic (As) ions may be doped into the substrate 100. Referring to FIG. 4D, a dielectric material may be deposited into the trenches 104 to fill the space between two adjacent fin structures 106. The dielectric material may include silicon oxide, silicon nitride, undoped silicate glass (USG), boro-silicate glass (BSG), tetraethyl orthosilicate (TEOS), or other suitable materials. An etching operation may be performed to etch the dielectric material in the trenches 104. As a result, top surfaces of the fin structures 106 may be higher than a top surface of the etched dielectric material. The dielectric material in the trenches 104 may form multiple insulating structures 108 between the

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fin structures 106. A portion of each of the fin structures 106 exposed from the insulating structures 108 may be referred to as a protruding fin.

FIG. 4E is a schematic plan view of FIG. 4D. In some embodiments, the fin structures 106 and the insulating structures 108 extend along the first direction D1. In some embodiments, the fin structures 106 and the insulating structures 108 are alternately arranged with each other along a second direction D2 perpendicular to the first direction D1. The fin structures 106 may be referred to as "oxide definition" or "oxide diffusion" (OD) regions. The term oxide definition region refers to an active region for a transistor, i.e., an area where a source, a drain, and a channel under a gate of the transistor are formed. The fin structures 106 containing doped ions may function as multiple active regions or well regions over the substrate 100 and are electrically isolated from one another by the insulating structures 108.

Referring to operation 205 of FIG. 2, multiple dummy gates 122 and 124 are formed on the fin structure 106, as shown in FIG. 5A. The dummy gates 122 and 124 may be formed by depositing a polycrystalline-silicon (poly-Si) layer or a polycrystalline-silicon-germanium (poly-SiGe) layer on the fin structure 106 using a chemical vapor deposition (CVD) operation, and/or other suitable methods. A lithographic operation and an etching operation may be used to remove portions of the polycrystalline-silicon layer to form the dummy gates 122 and 124 shown in FIG. 5A. In some embodiments, the dummy gates 122 and 124 are at least partially located over the fin structures 106; that is, the active regions or the oxide definition regions. The dummy gates 122 and 124 formed over the fin structure 106 may be used in subsequent operations. For example, in a "gate last" or "replacement gate" technique, the dummy gates 124 (also called sacrificial gates) are initially formed. After various operations are finished, portions or a whole of the dummy gates 124 will be removed and replaced with one or more materials of replacement gates. In some embodiments, the dummy gate 122 may be replaced with a CPODE structure. In some embodiments, a width W1 of the dummy gate 122 is equal to a width W2 of the dummy gate 124. In some other embodiments, the width W1 of the dummy gate 122 is different from the width W2 of the dummy gate 124.

FIG. 5B is a schematic plan view of FIG. 5A. In some embodiments, the dummy gate 122 is formed between two dummy gates 124. The dummy gates 122 and 124 may extend along the second direction D2 perpendicular to the fin structures 106. In some embodiments, portions of the fin structures 106 on one side of the dummy gate 122 are designed as a first active region 112A and other portions of the fin structures 106 on the other side of the dummy gate 122 are designed as a second active region 112B. The first active region 112A and the second active region 112B are two predetermined active regions and are connected at this stage, as shown in FIGS. 5A and 5B.

Referring to operation 207 of FIG. 2, a gate spacer 126 is deposited on each of the dummy gates 122 and 124, as shown in FIG. 6. The gate spacers 126 may be formed by depositing a dielectric material over the dummy gates 122, 124 and the fin structure 106 using CVD, physical vapor deposition (PVD), atomic layer deposition (ALD), and/or other suitable methods. In some embodiments, the dielectric material of the gate spacers 126 includes silicon oxide, silicon nitride, silicon oxynitride, silicon carbide, or other suitable materials. An etch operation may be used to remove horizontal portions of the dielectric material to leaving the vertical portions of the dielectric material on sidewalls of the

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dummy gates 122 and 124, thereby forming the gate spacers 126 as shown in FIG. 6. In some embodiments, the gate spacer 126 is formed along a sidewall of the dummy gate 122 or 124.

Referring to operation 209 of FIG. 2, multiple recesses R1 are formed in the fin structure 106, as shown in FIG. 7. The formation of the recesses R1 includes at least a lithographic operation and an etching operation. The etching operation may include a reactive ion etching (RIE), a wet etching, and/or other suitable methods. Portions of the fin structure 106 are removed by an isotropic or an anisotropic etching to form multiple recesses R1 adjacent to the gate spacers 126. A profile of the recesses R1 may be substantially triangle, trapezoid, pentagon, or hexagon shape depending on etching parameters of the etching operation. In some embodiments, the etching operation partially etches the fin structure 106 with the dummy gates 122, 124 and the gate spacers 126 serving as an etching mask. A portion of the recesses R1 may extend a predetermined depth of the fin structure 100. In some embodiments, the recess R1 is formed between the gate spacers 126 of adjacent dummy gates 122 and 124 or between adjacent two dummy gates 124. The recesses R1 may extend along the second direction D2.

Referring to operation 211 of FIG. 2, an epitaxial feature 130 is formed in the recess R1, as shown in FIG. 8. An epitaxial growth operation may be performed to form multiple epitaxial features 130 on the fin structure 106 and each epitaxial feature 130 is formed in each recess R1. The epitaxial growth operation may include selective epitaxy growth (SEG), molecular-beam epitaxy (MBE), vapor-phase epitaxy (VPE), liquid-phase epitaxy (LPE), and/or other suitable methods. In some embodiments, the epitaxial growth operation uses gaseous and/or liquid precursors, which interact with a composition of the fin structure 106. In some embodiments, the formation of the epitaxial features 130 includes doping of N-type or P-type dopants and hence the epitaxial features 130 are conductive. In some embodiments, the epitaxial features 130 are in-situ doped or undoped during the epitaxial growth operation. For example, when undoped during the epitaxial growth operation, the epitaxial features 130 are doped in a subsequent operation. The epitaxial features 130 may be further exposed to annealing, such as a rapid thermal annealing (RTA) operation, to diffuse dopants within the epitaxial features 130.

A silicide (not shown) such as cobalt-silicide (CoSi), nickel silicide (NiSi) or tungsten silicide (WSi) may be formed on a top surface of each epitaxial feature 130. The silicide may include a self-aligned silicide (salicide). In some embodiments, the dummy gate 124 is between two consecutive epitaxial features 130. In some embodiments, the two consecutive epitaxial features 130 on one side of the dummy gate 122 may function as the source/drain features of a transistor formed in subsequent operations. In some embodiments, the two consecutive epitaxial features 130 on the other side of the dummy gate 122 may function as the source/drain features of another transistor formed in subsequent operations.

Referring to operation 213 of FIG. 2, an etch stop layer 140 is formed on the gate spacer 126 and the epitaxial feature 130, as shown in FIG. 9. The etch stop layer 140 may be formed by depositing a dielectric material onto the gate spacer 126 and the epitaxial feature 130 using CVD, PVD, ALD, and/or other suitable methods. In some embodiments, the dielectric material of the etch stop layer 140 includes silicon oxide, silicon nitride, silicon oxynitride, silicon carbide, silicon oxycarbide, or other suitable materials. An etch operation may be used to remove portions of the dielectric

material to form multiple etch stop layers **140** shown in FIG. **9**. In some embodiments, the etch stop layer **140** may be formed along a sidewall of the gate spacer **126**. As such, the dummy gates **122** and **124** are surrounded by the etch stop layers **140**. In some embodiments, the epitaxial feature **130** is completely covered by the etch stop layer **140**. In some other embodiments, the epitaxial feature **130** is partially covered by the etch stop layer **140** when the etch operation removes a portion of the etch stop layer **140** over the epitaxial feature **130**. The etch stop layer **140** may be a contact etch stop layer (CESL) used for subsequent operations. The CESL is formed on sidewalls of the dummy gates **122**, **124** and on the epitaxial features **130**.

Referring to operation **215** of FIG. **2**, an inter-layer dielectric (ILD) layer **150** is formed on the etch stop layer **140** and the epitaxial feature **130**, as shown in FIG. **10**. The ILD layer **150** may be formed by depositing an insulating material onto the dummy gates **122**, **124** and the etch stop layers **140** using spin-on coating, CVD, PVD, ALD, and/or other suitable methods. In some embodiments, the insulating material of the ILD layer **150** includes silicon oxide, silicon nitride, undoped silicate glass (USG), borosilicate glass (BSG), a low-k material, tetraethyl orthosilicate (TEOS), or other suitable materials. After the space between the dummy gates **122** and **124** and between two adjacent dummy gates **124** are completely filled with the insulating material, a CMP operation may be used to remove the insulating material over the dummy gates **122** and **124**. As such, multiple ILD layer **150** are formed and alternately arranged with the dummy gates **122**, **124**. After the CMP operation, top surfaces of the dummy gates **122** and **124** are exposed. A top surface of the ILD layer **150** is substantially coplanar with a top surface of dummy gates **122** and **124**.

Referring to operation **217** of FIG. **2**, a first patterned photoresist layer **154** is formed on the dummy gates **122** and **124**, as shown in FIGS. **11A** and **11B**. Referring to FIG. **11A**, a first photoresist layer **152** is coated to cover the etch stop layer **140**, the ILD layer **150** and the dummy gates **122** and **124**. Subsequently, the first photoresist layer **152** is exposed to a first radiation P1 via a photomask M1. After exposure, portions of the first photoresist layer **152** are developed, and the first patterned photoresist layer **154** is formed, as shown in FIG. **11B**. In some embodiments, the first patterned photoresist layer **154** includes an opening O1A that exposes the dummy gate **122**.

Referring to operation **219** of FIG. **2**, a trench T1 is formed to penetrate the dummy gate **122**, as shown in FIGS. **12A** and **12B**. In some embodiments, the formation of the trench T1 includes multiple etching operations such as continuous dry etching or RIE using the first patterned photoresist layer **154** as an etching mask. In some embodiments, the etching operations vertically remove the dummy gate **122** and portions of the fin structure **106** and the substrate **100** underneath the dummy gate **122**. In such embodiments, the trench T1 penetrates the fin structure **106** and extends to a depth of the substrate **100**.

FIG. **12B** is a schematic plan view of FIG. **12A**. Some elements are omitted in FIG. **12B** for clarity of the drawing. Comparing FIGS. **5B** and **12B**, since the fin structure **106** is penetrated by the trench T1, the first active region **112A** may be separated from the second active region **112B** by the trench T1. In other words, the trench T1 separates one active region into two active regions such as the first active region **112A** and the second active region **112B**. The trench T1 may be used to form a CPODE structure. In some embodiments, the trench T1 cuts through at least two neighboring fins **130** (i.e., oxide definition regions). After the dummy gate **122**

and the fin structure **106** are penetrated by the trench T1, the substrate **100** may be exposed by the trench T1, as shown in FIG. **12B**.

Referring to operation **221** of FIG. **2**, the trench T1 is filled with multiple dielectric layers, as shown in FIGS. **13A** to **13D**. In some embodiments, the dielectric layers include a multilayer structure formed of one or more dielectric layers, such as an oxide-nitride-oxide (ONO) structure. The formation of the dielectric layers includes one or more deposition, lithographic and etching operations.

Referring to FIG. **13A**, a first dielectric layer **162** is deposited over the etch stop layer **140**, the ILD layer **150** and the dummy gates **124**, and along sidewalls of the trench T1. The first dielectric layer **162** may be formed using CVD, ALD, and/or other suitable methods. The first dielectric material **162** may include silicon oxide or other suitable materials. The first dielectric material **162** may be deposited along sidewalls of the gate spacer **126**, the fin structure **106** and the substrate **100** in a conformal manner. In some embodiments, the first dielectric material **162** has a thickness D1 of about 10 angstroms (Å) to about 30 Å.

Referring to FIG. **13B**, a second dielectric layer **164** is deposited over the first dielectric layer **162** using CVD, ALD, and/or other suitable methods. The second dielectric layer **164** may include silicon nitride or other suitable materials. The second dielectric layer **164** may be deposited over the first dielectric layer **162** in a conformal manner. In some embodiments, the second dielectric layer **164** has a thickness D2 of about 30 Å to about 50 Å.

Referring to FIG. **13C**, a third dielectric layer **166** is deposited over the second dielectric layer **164** using CVD, ALD, and/or other suitable methods. The third dielectric layer **166** may include silicon oxide or other suitable materials. The third dielectric layer **166** may be deposited over the second dielectric layer **164** in a conformal manner. In some embodiments, the third dielectric layer **166** has a thickness D3 of about 20 Å to about 40 Å. The first dielectric layer **162**, the second dielectric layer **164** and the third dielectric layer **166** may form a dielectric stack **160** in the trench T1. In some embodiments, the dielectric stack **160** substantially fills the trench T1. In some other embodiments, a void is left in the trench T1 after the dielectric stack **160** is formed in the trench T1. The first dielectric layer **162** and the second dielectric layer **164** may have a U-shape profile, but the configuration thereof is not limited herein. In some embodiments, the dummy gate **124**, the gate spacer **126**, the etch stop layer **140** and the ILD layer **150** are covered by the dielectric stack **160**. In some embodiments, the dielectric stack **160** includes a lower portion **161**, an upper portion **163** over the lower portion **161** and pair of horizontal portions **165** on opposite sides of the upper portion **163**.

Referring to FIG. **13D**, a cap layer **168** is deposited over the dielectric stack **160** using spin-on coating, CVD, ALD, and/or other suitable methods. The cap layer **168** may include silicon nitride, silicon oxynitride, or other suitable materials. In some embodiments, the cap layer **168** does not include any silicon oxide. An etch operation or a CMP operation may be used to planarize the cap layer **168** to make the cap layer **168** have a flat top surface.

Referring to operation **223** of FIG. **2**, a second patterned photoresist layer **174** is formed over the dielectric layers **162**, **164**, **166** and the cap layer **168**, as shown in FIGS. **14A** and **14B**. Referring to FIG. **14A**, a second photoresist layer **172** is deposited over the cap layer **168**. Subsequently, the second photoresist layer **172** is exposed to a second radiation P2 via the photomask M1. After exposure, portions of the second photoresist layer **172** are developed, and the second

patterned photoresist layer **174** is formed, as shown in FIG. **14B**. In some embodiments, the second patterned photoresist layer **174** includes an opening **O2A** that exposes a top surface of the cap layer **168** and corresponds to the trench **T1**.

Referring to operation **225** of FIG. **2**, a trench **T2** is formed to penetrate the cap layer **168**, as shown in FIG. **15A**. In some embodiments, the formation of the trench **T2** includes an etching operation such as dry etching or RIE using the second patterned photoresist layer **174** as an etching mask. In some embodiments, the etching operation vertically removes a portion of the dielectric stack **160** and a portion of the cap layer **168**. The dielectric stack **160** may be etched through the opening **O2A** until the trench **T2** with a predetermined depth **Z1** is formed. In some embodiments, the upper portion **163** of the dielectric stack **160** is substantially removed during the etching operation. After the upper portion **163** is removed, the lower portion **161** is left in the trench **T2** and the horizontal portions **165** are left over the dummy gates **124**. The lower portion **161** may have a top surface having a concave upward shape. The lower portion **161** has a vertical profile and may be referred to as a CPODE structure **161**. The CPODE structure **161** may be an isolating structure for providing electrical isolation between neighboring features. In some embodiments, multiple CPODE structures **161** are disposed over the substrate **100**. In some embodiments, a top surface **S1** of the CPODE structure **161** is lower than a top surface of the dummy gate **124**, the gate spacer **126** or the ILD layer **150**. In some embodiments, the top surface **S1** of the CPODE structure **161** (the lower portion **161**) is higher than the topmost surface **S2** of the fin structure **106** by about 30 Å to about 60 Å and lower than a top surface of the dummy gate **124**. Since the trench **T1** penetrates the fin structure **106** and extends into the substrate **100**, the first active region **112A** and the second active region **112B** may be separated by the CPODE structure **161**.

FIG. **15B** shows an enlarged view of the top surface **S1** of the CPODE structure **161** in FIG. **15A**. In some other embodiments, the top surface **S1** of the CPODE structure **161** is substantially a concave surface. Due to a difference of etching rates among the first, second and third dielectric materials **162**, **164** and **166**, the CPODE structure **161** may have a round profile on its top portion after the etching operation.

Referring to operation **227** of FIG. **2**, a protective layer **169** is formed on the cap layer **168**, as shown in FIG. **16**. The protective layer **169** may be formed by depositing an insulating material over the cap layer **168** and within the trench **T2** using spin-on coating, CVD, ALD, and/or other suitable methods. The insulating material of the protective layer **169** may include silicon nitride, silicon oxynitride, silicon carbon nitride, or other suitable materials. In some embodiments, the protective layer **169** does not include any silicon oxide. In some embodiments, the protective layer **169** in the trench **T2** is directly disposed on and completely covering the lower portion **161**.

Referring to operation **229** of FIG. **2**, portions of the cap layer **168** and the protective layer **169** are removed, as shown in FIGS. **17A** and **17B**. Referring to FIG. **17A**, a removal operation such as a CMP operation or an etching operation may be used to remove excess portions of the cap layer **168** and the protective layer **169** above a topmost surface of the horizontal portion **165**. After the removal operation, the protective layer **169** is formed only directly over the CPODE structure **161** and the horizontal portion

165 is exposed. In some embodiments, top surfaces of the protective layer **169** and the horizontal portions **165** are substantially coplanar.

FIG. **17B** shows an enlarged view of the protective layer **169** after the removal operation in FIG. **17A**. In some embodiments, a width **L1** of the protective layer **169** after the removal operation is about 15 nanometers (nm) to about 40 nm, and a height **L2** of the protective layer **169** after the removal operation is about 10 nm to about 100 nm. In some embodiments, the protective layer **169** has rounded corners around its bottom portions due to the concave top surface **S1** of the CPODE structure **161**. In some embodiments, the CPODE structure **161** has a substantially triangular portion over the top surface **S1** as a protruding portion **161T**, as shown in FIG. **17B**. The protruding portion **161T** may have a maximal thickness **P1** of about 2 nm to about 10 nm.

Referring to operation **231** of FIG. **2**, the dummy gates **124** are removed to form multiple through holes **H1**, as shown in FIGS. **18A** and **18B**. Referring to FIG. **18A**, the horizontal portions **165** are partially etched to respectively expose the underlying dummy gates **124**. Referring to FIG. **18B**, the exposed dummy gates **124** may be etched using an etchant selective to polysilicon. In some embodiments, the etching operation of the dummy gates **124** includes a wet etching by using a solution such as NH_4OH , dilute HF , and/or other suitable etchant, or a dry etching by using a gas such as fluorine-based and/or chlorine-based etchants for removing the dummy gates **124**. The gate spacers **126** and the horizontal portions **165** may function as an etchant mask and are not consumed (or substantially not consumed) during the removal of the dummy gates **124**. After the dummy gates **124** are removed, the through holes **H1** are formed. In some embodiments, each through hole **H1** is defined by the corresponding gate spacer **126**. A width **W3** of the through hole **H1** can be adjusted based on the width **W2** of the dummy gate **124**. In some embodiments, the width **W3** of the through hole **H1** is in range of about 20 nm to about 50 nm. The through holes **H1** over the fin structure **106** may extend along the first direction **D1**.

Referring to operation **233** of FIG. **2**, a gate structure **180** is formed in each of the through holes **H1**, as shown in FIG. **19**. The gate structure **180** may be a gate stack that includes multiple layers. The formation of the gate structure **180** may include multiple deposition operations and one or more etching operations. For example, various dielectric materials and conductive materials may be deposited in sequence in a blanket manner, followed by a patterning operation to form a desired profile of the gate structure **180**. In some embodiments, a gate dielectric layer **182** is formed in through hole **H1** on the fin structure **106** using a thermal oxidation operation or a CVD operation. The gate dielectric layer **182** may include one or more dielectric materials, such as silicon oxide, silicon nitride, silicon oxynitride, or other suitable materials. The gate dielectric layer **182** may have a planar profile.

In some embodiments, a high-k material layer **184** is formed on the gate dielectric layer **182** using CVD, PVD, ALD, and/or other suitable methods. The high-k material layer **184** may include one or more dielectric materials with high dielectric constants higher than that of silicon oxide, such as Al_2O_3 , HfO_2 , ZrO_2 , HfZrO , HfO_xN_y , ZrO_xN_y , HfSi_xO_y , ZrSi_xO_y , $\text{HfSi}_x\text{O}_y\text{N}_z$, $\text{ZrSi}_x\text{O}_y\text{N}_z$, TiO_2 , Ta_2O_5 , Ta_2O_3 , CeO_2 , WO_3 , Y_2O_3 , LaAlO_3 , $\text{Ba}_{1-x}\text{Sr}_x\text{TiO}_3$, PbTiO_3 , BaTiO_3 (BTO), SrTiO_3 (STO), BaSrTiO_3 (BST), PbZrO_3 , lead-strontium-titanate (PST), lead-zinc-niobate (PZN), lead-zirconate-titanate (PZT), lead-magnesium-niobium (PMN), yttria-stabilized zirconia (YSZ), $\text{ZnO}/\text{Ag}/\text{ZnO}$

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(ZAZ), a combination thereof, or other suitable materials. The high-k material layer **184** may have a U-shape or a planar profile.

In some embodiments, a gate electrode **186** is formed on the high-k material layer **184** using an electroplating operation or a PVD operation. The gate electrode **186** may include one or more conductive materials, such as tungsten (W), copper (Cu), cobalt (Co), aluminum (Al), nickel (Ni), tantalum (Ta), titanium (Ti), molybdenum (Mo), palladium (Pd), platinum (Pt), ruthenium (Ru), iridium (Ir) silver (Ag), gold (Au), titanium nitride (TiN), tantalum nitride (Ta₃N₅), a combination thereof, or other suitable materials. In one embodiment, when the high-k material layer **184** has a U-shape profile, the conductive material is deposited in a cavity surrounded by the high-k material layer **184**. In another embodiment, when the high-k material layer **184** has a planar profile, the conductive material is deposited to completely cover the high-k material layer **184**.

In some embodiments, the gate dielectric layer **182**, the high-k material layer **184** and the gate electrode **186** are sequentially deposited in the through hole **H1** to form the gate structure **180**. However, details of the formation of the gate structure **180** are not limited herein. Multiple gate structures **180** may be formed on the fin structure **106** and each of the gate structures **180** may be surrounded by the corresponding gate spacers **126**. In some embodiments, a height of the gate structure **180** is less than a height of the gate spacer **126**. In some other embodiments, a height of the gate structure **180** is substantially equal to a height of the gate spacer **126**.

Still referring to FIG. **19**, a gate structure **180** and its adjacent epitaxial features **130** on one side of the CPODE structure **161** may form a transistor **T10**, and another gate structure **180** and its adjacent pair of epitaxial features **130** on the other side of the CPODE structure **161** may form a transistor **T20**. The gate structure **180** may function as a gate terminal of the transistor **T10** or **T20**, and the pair of epitaxial features **130** may function as source/drain terminals of the transistor **T10** or **T20**. The transistors **T10** and **T20** may be field-effect transistor (FET) devices. In some embodiments, the transistors **T10** and **T20** are planar-type FET devices, fin-type FET devices, gate-all-around (GAA) FET devices, nanosheet FET devices or nanowire FET devices, but the configuration of the transistors **T10** and **T20** is not limited thereto. In some embodiments, the transistors **T10** and **T20** are N-type MOSFET (NMOS) when the epitaxial features **130** include N-type dopants. In some other embodiments, the transistors **T10** and **T20** are P-type MOSFET (PMOS) when the epitaxial features **130** include P-type dopants. Still in some embodiments, the transistor **T10** is an NMOS transistor while the transistor **T20** is a PMOS transistor, or the transistor **T10** is a PMOS transistor while the transistor **T20** is an NMOS transistor.

Comparing FIGS. **10** and **19**, each of the dummy gates **124** is replaced by a corresponding gate structure **180**. The conversion of a dummy gate (such as the dummy gate **124**) into a functional gate (such as the gate structure **180**) uses a technique referred to as "replacement polysilicon gate (RPG)". The dummy gate **124** does not function as a gate terminal of a transistor, but is used as a sacrificial structure for forming the transistor. During the formation of the transistors **T10** and **T20**, the dummy gates **124** are replaced by the functional gate structures **180**. After the formation of the gate structures **180**, the transistors **T10** and **T20** are completed. Thus, the RPG technique may also be called a "gate last" technique.

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Referring to operation **235** of FIG. **2**, a hard mask **190** is formed on the gate structure **180** and the gate spacer **126**, as shown in FIG. **20**. The formation of the hard mask **190** includes depositing silicon nitride to completely cover the gate structure **180**. A CMP operation may be used to remove the excess material of the hard mask **190** over the horizontal portions **165**. As such, a top surface of the hard mask **190** is substantially coplanar with top surfaces of the horizontal portions **165** or the protective layer **169**. The operation steps for forming the features discussed above with reference to FIG. **3** to FIG. **20** may be collectively referred to as the FEOL stage. In the following, descriptions of operation steps in the MEOL stage and BEOL stage are provided.

In the MEOL stage, a conductive structure such as a conductive contact or a conductive via is formed over the substrate. The conductive structure may be used to electrically connect a gate terminal or a source/drain terminal of a transistor to one or more conductive layers formed in the BEOL stage.

Referring to operation **237** of FIG. **2**, a conductive via **192** is formed on the epitaxial feature **130**, as shown in FIGS. **21** and **22**. Referring to FIG. **21**, a contact hole **C1** is formed to expose the epitaxial feature **130**. In some embodiments, the formation of the contact hole **C1** does not include a lithographic operation. The hard mask **190** and the protective layer **169** may be directly used as an etching mask for protecting the underlying layers. An etching operation such as RIE may be used to vertically etch the horizontal portion **165**, the ILD layer **150** and the etch stop layer **140** until the epitaxial feature **130** is exposed. Multiple contact holes **C1** may be formed and each contact hole **C1** exposes the underlying epitaxial feature **130**.

In some embodiments, the etchant used in the formation of the contact holes **C1** may be reactive to the horizontal portions **165** since the horizontal portions **165** are to be etched as well. Therefore, if the CPODE structure **161**, which has the same composition as the horizontal portions **165**, is exposed during the formation of the contact holes **C1**, the CPODE structure **161** may be damaged, thereby degrading its function of preventing a leakage current between the two transistors **T10** and **T20**. In some embodiments, the protective layer **169** isolates the CPODE structure **161** from the etchant used in the formation of the contact holes **C1** and prevents the CPODE structure **161** from being consumed or damaged by the etchant. As a result, the CPODE structure **161** substantially remains intact after the contact holes **C1** are formed. The transistors **T10** and **T20** can be separated by the CPODE structure **161**. Since the CPODE structure **161** remains intact, after the contact vias **192** are subsequently formed in adjacent contact holes **C1** on two sides of the CPODE structure **161**, electrical bridging between the two transistors **T10** and **T20** may be reduced or prevented. When the transistors **T10** and **T20** are different types of transistors, a difference between threshold voltages (V_{th}) of the NMOS and PMOS transistors is a critical issue in the operation of low voltage circuits. Since the CPODE structure **161** includes multiple dielectric materials arranged in a stacked manner, the P/N threshold voltages can be significantly balanced. In some embodiments, balancing the P/N threshold voltages reduces a lowest required supply voltage by about 0.15 volts (V) to about 0.3 V.

Referring to FIG. **22**, a conductive material is deposited into the contact hole **C1** using electroplating, PVD, CVD, and/or other suitable methods. The conductive material may include one or more conductive materials, such as tungsten (W), copper (Cu), cobalt (Co), aluminum (Al), nickel (Ni), tantalum (Ta), titanium (Ti), molybdenum (Mo), palladium

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(Pd), platinum (Pt), ruthenium (Ru), iridium (Ir) silver (Ag), gold (Au), or other suitable materials. A CMP operation may be used to remove the conductive material over the hard mask **190**. As such, the conductive via **192** is formed in the contact hole **C1** and on the epitaxial feature **130**. The conductive via **192** may extend through the ILD layer **150**. In some embodiments, the conductive via **192** is electrically coupled to the underlying epitaxial feature **130**. The conductive via **192** may be referred to herein as a metal-to-drain (MD) feature that connects a doped region (i.e., the epitaxial feature **130**) in the fin structure **106** to a conductive layer overlying the doped region.

Referring to operation **239** of FIG. **2**, a conductive via **194** is formed on the gate structure **180**, as shown in FIGS. **23** and **24**. Referring to FIG. **23**, a contact hole **C2** is formed to penetrate the hard mask **190**. The formation of the contact hole **C2** includes at least a lithographic operation and an etching operation such as dry etching or RIE. In some embodiments, the etching operation removes a portion of the hard mask **190** to expose the gate electrode **186**. Multiple contact holes **C2** may be formed and each contact hole **C2** exposes the underlying gate electrode **186**.

Referring to FIG. **24**, a conductive material is deposited into the contact hole **C2** using electroplating, PVD, CVD, and/or other suitable methods. The conductive material may be the same material as that of the conductive via **192**. A CMP operation may be used to remove the conductive material over the hard mask **190**. As such, the conductive via **194** is formed in the contact hole **C2** and on the gate electrode **186**. The conductive via **194** may extend through the hard mask **190**. In some embodiments, the conductive via **194** is electrically coupled to the underlying gate electrode **186**. The conductive via **194** may be referred to herein as a via-to-gate (VG) feature that connects a gate terminal (i.e., the gate electrode **186**) of the gate structure **180** to a conductive layer overlying the gate terminal.

In the BEOL stage, multiple dielectric and conductive features are formed. Referring to operation **241** of FIG. **2**, an interconnect layer **250** is formed over the conductive vias **192** and **194**, as shown in FIG. **25**. The interconnect layer **250** includes a metal line layer **252** embedded in an ILD layer **254**. The formation of the interconnect layer **250** includes one or more deposition, lithographic and etching operations. The metal line layer **252** may include one or more conductive materials, such as tungsten (W), copper (Cu), cobalt (Co), aluminum (Al), nickel (Ni), tantalum (Ta) or titanium (Ti). The ILD layer **254** may include a low dielectric constant (k value less than about 3.0) or an extreme low dielectric constant (k value less than about 2.5) material. For example, the ILD layer **254** may be made of silicon oxide, borophosphosilicate glass (BPSG), TEOS, spin-on glass (SOG), undoped silicate glass (USG), fluorinated silicate glass (FSG), high-density plasma (HDP) oxide, or plasma-enhanced TEOS (PETEOS). A CMP operation may be used to planarize a top surface of the interconnect layer **250**. In some embodiments, the metal line layer **252** is disposed over and electrically coupled to the conductive vias **192** and **194**. The metal line layer **252** may be referred to herein as an M_0 metal line layer that electrically connects various features over the substrate **100**, such as the transistor **T10** and the transistor **T20**, to overlying features.

Referring to operation **243** of FIG. **2**, a metallization layer **300** is formed over the interconnect layer **250**, as shown in FIG. **26**. The formation of the metallization layer **300** may include a series of deposition, lithographic, etching and planarization operations. The metallization layer **300** may be formed of a stack of interconnected metal line layers (wires)

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and metal via layers. For example, the metal line layers $M_1, \dots, M_{n-1}, M_n$ (the integer n denotes the layer index) are formed over one another and include horizontally extending metal lines. The metal via layers $V_0, \dots, V_{n-1}$ (the integer n denotes the layer index) are formed over one another and include vertically extending metal lines. A metal via layer V_n is interposed between an underlying metal line layer M_n and an overlying metal line layer M_{n+1} and electrically connects the metal line layer M_n and M_{n+1} . People having ordinary skill in the art can understand that the numbers provided for the metal line layers M_n and metal via layers V_n described herein are not limiting. The numbers and configurations of the metallization layer **300** can be adjusted on demand according to the design requirement. The metallization layer **300** may be referred to as a stacked routing structure formed in the BEOL stage. As such, the semiconductor structure **10** is formed.

One aspect of the present disclosure provides a method of manufacturing a semiconductor structure. The method includes providing a substrate; forming a fin structure on the substrate; forming a first dummy gate on the fin structure; forming a trench to penetrate the first dummy gate and the fin structure; forming a dielectric stack in the trench; removing a top portion of the dielectric stack in the trench to leave a lower portion of the dielectric stack in the trench; and forming a protective layer in the trench and directly on the lower portion of the dielectric stack.

One aspect of the present disclosure provides another method of manufacturing a semiconductor structure. The method includes forming a fin structure on a substrate; forming a first dummy gate, a second dummy gate and a third dummy gate over the fin structure, wherein the second dummy gate is between the first dummy gate and the third dummy gate; forming a trench to penetrate the second dummy gate and the fin structure; forming a dielectric stack in the trench and over the first dummy gate and the third dummy gate; forming a protective layer in the trench over the dielectric stack.

Another aspect of the present disclosure provides a semiconductor structure. The semiconductor structure includes a substrate; a fin structure on the substrate; a first transistor on the fin structure, the first transistor including a first gate structure and a first source/drain structure; a second transistor on the fin structure and adjacent to the first transistor, the second transistor including a second gate structure and a second source/drain structure; a first conductive via disposed over the first source/drain structure; a second conductive via disposed over the second source/drain structure; an insulating structure penetrating the fin structure; and a protective layer directly over the insulating structure, wherein the insulating structure and the protective layer separate the first conductive via and the second conductive via.

The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

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What is claimed is:

1. A method of manufacturing a semiconductor structure, comprising:
 - providing a substrate;
 - forming a fin structure on the substrate;
 - forming a first dummy gate on the fin structure;
 - forming a trench to penetrate the first dummy gate and the fin structure;
 - forming a dielectric stack in the trench;
 - removing a top portion of the dielectric stack in the trench to leave a lower portion of the dielectric stack in the trench; and
 - forming a protective layer in the trench and directly on the lower portion of the dielectric stack, wherein the forming of the protective layer includes:
 - depositing a cap layer over the dielectric stack;
 - patterning the cap layer and the dielectric stack to form an opening aligned with the trench; and
 - filling the opening with the protective layer.
2. The method of claim 1, further comprising:
 - forming a second dummy gate and a third dummy gate on opposite sides of the first dummy gate after the forming of the fin structure; and
 - forming a first epitaxial feature on opposite sides of the second dummy gate and a second epitaxial feature on opposite sides of the third dummy gate.
3. The method of claim 2, wherein after the removing of the top portion of the dielectric stack in the trench, a top surface of the lower portion is lower than a top surface of the second dummy gate or the third dummy gate.
4. The method of claim 3, wherein the forming of the trench includes:
 - coating a photoresist layer over the second dummy gate and the third dummy gate;
 - exposing the photoresist layer to radiation via a photo-mask;
 - developing the photoresist layer to form a patterned photoresist layer including a first opening over the first dummy gate; and
 - etching the first dummy gate, the fin structure and the substrate by using the patterned photoresist layer as an etching mask.
5. The method of claim 2, further comprising:
 - forming an interlayer dielectric (ILD) layer on the first epitaxial feature and the second epitaxial feature; and
 - forming a plurality of contact holes by etching the ILD layer until the first epitaxial feature and the second epitaxial feature are exposed, wherein the protective layer is used as an etching mask during the forming of the plurality of contact holes.
6. The method of claim 1, wherein the forming of the dielectric stack includes:
 - conformally depositing a first dielectric layer containing silicon oxide into the trench;
 - conformally depositing a second dielectric layer containing silicon nitride over the first dielectric layer; and
 - depositing a third dielectric layer containing silicon oxide over the second dielectric layer.
7. The method of claim 1, wherein the protective layer includes silicon nitride, silicon oxynitride or silicon carbon nitride.
8. The method of claim 1, wherein the dielectric stack in the trench remains intact after the forming of the plurality of contact holes.
9. The method of claim 1, wherein the protective layer is made of silicon nitride or silicon carbon nitride.

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10. A method of manufacturing a semiconductor structure, comprising:
 - forming a fin structure on a substrate;
 - forming a first dummy gate, a second dummy gate and a third dummy gate over the fin structure, wherein the second dummy gate is between the first dummy gate and the third dummy gate;
 - forming a first epitaxial feature and a second epitaxial feature on the fin structure, wherein the first epitaxial feature is between the first dummy gate and the second dummy gate, and the second epitaxial feature is between the second dummy gate and the third dummy gate;
 - forming an interlayer dielectric (ILD) layer on the first epitaxial feature and the second epitaxial feature;
 - forming a trench to penetrate the second dummy gate and the fin structure;
 - forming a dielectric stack in the trench and over the first dummy gate and the third dummy gate;
 - depositing a cap layer over the dielectric stack;
 - patterning the cap layer and the dielectric stack to form an opening aligned with the trench; and
 - filling the opening with a protective layer.
11. The method of claim 10, wherein after the forming of the protective layer, further comprising:
 - replacing the first dummy gate with a first functional gate and replacing the third dummy gate with a second functional gate, wherein each of the first functional gate and the second functional gate includes a gate dielectric layer, a high-k material layer over the gate dielectric layer and a gate electrode over the high-k material layer.
12. The method of claim 11, further comprising:
 - depositing a hard mask over the first and second functional gates;
 - etching the ILD layer by using the hard mask and the protective layer as an etching mask until the first epitaxial feature and the second epitaxial feature are exposed; and
 - depositing a conductive material on the first epitaxial feature and the second epitaxial feature to form a first conductive via and a second conductive via, respectively.
13. The method of claim 12, wherein the protective layer prevents a lower portion of the dielectric stack in the trench from being consumed during the etching of the ILD layer.
14. The method of claim 12, wherein the protective layer prevents a lower portion of the dielectric stack in the trench from exposure to an etchant used during the etching of the ILD layer.
15. The method of claim 10, prior to the forming of the protective layer, further comprising removing a top portion of the dielectric stack in the trench to leave a lower portion of the dielectric stack in the trench, wherein after the removing of the top portion of the dielectric stack in the trench, the lower portion of the dielectric stack includes a top surface having a concave upward shape.
16. A method of manufacturing a semiconductor structure, comprising:
 - forming a fin structure on a substrate;
 - forming a first transistor on the fin structure, wherein the first transistor includes a first gate structure and a first source/drain structure;
 - forming a second transistor on the fin structure and adjacent to the first transistor, wherein the second transistor includes a second gate structure and a second source/drain structure;

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forming a first conductive via over the first source/drain structure and a second conductive via over the second source/drain structure;

recessing a portion of the fin structure to form a trench, wherein the trench is between the first transistor and the second transistor;

filling the trench with a dielectric stack;

depositing a cap layer over the dielectric stack;

patterning the cap layer and the dielectric stack to form an opening aligned with the trench; and

filling the opening with a protective layer.

17. The method of claim **16**, wherein the dielectric stack and the protective layer separate the first conductive via and the second conductive via.

18. The method of claim **16**, wherein the trench extends into the substrate.

19. The method of claim **16**, wherein a top surface of the dielectric stack is higher than a top surface of the fin structure.

20. The method of claim **16**, wherein the dielectric stack includes a first oxide layer, a nitride layer over the first oxide layer and a second oxide layer over the nitride layer.

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